CONTINUATION OF
SECTION B – SUPPLIES OR SERVICES AND PRICES

SAN JUAN LATERAL – TURNOUT BUILDINGS
NAVAJO-GALLUP WATER SUPPLY PROJECT, NEW MEXICO

B.1 PRICE SCHEDULES

(a) Offers will be considered for award on the following Price Schedules, but no offer will be considered for award on only part of the Price Schedules. Offers are subject to the terms and conditions of this solicitation.

(b) Quantities in Price Schedules are estimated quantities for comparison of offers only. Except as provided with the provision at H.2, Variation in Estimated Quantity, no claim shall be made against Government for overruns or underruns. Contract clause at H.2, Variation in Estimated Quantity does not apply to lump sum items.

(c) See contract clause at WBR 1452.232-81, Payment for Mobilization and Preparatory Work, for CLINs 1-1 and 2-1.

(d) Definitions:

(1) CLIN – Contract Line Item Number

(e) Note: In accordance with Section 7-9-3.3 of New Mexico Statutes Annotated (NMSA) 1978 and Section 402 of Navajo Nation's Business Activity Tax Statute, it is Contractor's responsibility to ensure that all applicable New Mexico Gross Receipts Tax and Navajo Nation Gross Receipts Tax are included in their price proposal.

(f) In accordance with FAR 52.232-5 (Payments under Fixed-Price Construction Contracts) (b)(2), payment may be approved for materials delivered to Contractor at locations other than the site if Contractor obtains CO approval prior to making arrangements for obtaining the materials. Contractor shall also be required to complete a "Materials on Hand Certificate" which will certify:

(1) Materials described for use on identified contract.

(2) Materials are clearly marked to identify that they belong to contract cited.

(3) Full title for those materials vested in Prime Contractor sited.

(4) Materials are free of liens and encumbrances.

(5) Materials are properly stored, secured, and protected against loss or damage.

PRICE SCHEDULE 1 (Sanostee)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
1-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
~~***1-2***~~	~~***01 31 30***~~	~~***Deleted Contract Document Management System (CDMS)***~~	~~***For the lump sum of***~~		~~***$***~~
1-3	01 56 15	Utility Crossing Investigations	For the lump sum of		$
1-4	01 78 30	As-built Drawings	For the lump sum of		$10,000
1-5	01 78 30	Final Drawings	For the lump sum of		$5,000
1-6	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
1-7	13 34 00	Turnout Buildings	For the lump sum of		$
1-8	25 00 01	System Control and Monitoring	For the lump sum of		$
1-9	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
1-10	26 05 33	Electrical Conduit	100 ft.	$	$
1-11	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
1-12	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
1-13	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
1-14	27 30 01	Telephone System	For the lump sum of		$
1-15	28 10 01	Security and Surveillance System	For the lump sum of		$
1-16	31 02 10	Water for Dust Abatement	15 M (1,000) Gal	$	$
1-17	31 02 30	Dust Palliative	0.19 acre	$	$

PRICE SCHEDULE 1 (Sanostee)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
1-18	31 37 00	Riprap	50 yd^3	$	$
1-19	32 15 10	Gravel Surfacing	86 yd^3	$	$
1-20	32 31 19	Steel Ornamental Fence	240 ft.	$	$
1-21	32 92 20	Seeding	0.06 acre	$	$
1-22	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
1-23	35 21 95	Steel Manifold Piping	For the lump sum of		$
1-24	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 1 $_____

PRICE SCHEDULE 2 (Newcomb)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
2-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
2-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
2-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
2-4	01 78 30	Final Drawings	For the lump sum of		$5,000
2-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
2-6	13 34 00	Turnout Buildings	For the lump sum of		$
2-7	25 00 01	System Control and Monitoring	For the lump sum of		$
2-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
2-9	26 05 33	Electrical Conduit	280 ft.	$	$

PRICE SCHEDULE 2 (Newcomb)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
2-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
2-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
2-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
2-13	27 30 01	Telephone System	For the lump sum of		$
2-14	28 10 01	Security and Surveillance System	For the lump sum of		$
2-15	31 02 10	Water for Dust Abatement	9.2 M (1,000) Gal	$	$
2-16	31 02 30	Dust Palliative	0.13 acre	$	$
2-17	***31 23 50***	***Deleted***			
2-19	31 37 00	Riprap	32 yd^3	$	$
2-20	32 15 10	Gravel Surfacing	49 yd^3	$	$
2-21	32 31 19	Steel Ornamental Fence	194 ft.	$	$
2-22	32 92 20	Seeding	0.05 acre	$	$
2-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
2-24	35 21 95	Steel Manifold Piping	For the lump sum of		$
2-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 2 $_____

PRICE SCHEDULE 3 (Naschitti)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
3-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
3-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
3-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
3-4	01 78 30	Final Drawings	For the lump sum of		$5,000
3-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
3-6	13 34 00	Turnout Buildings	For the lump sum of		$
3-7	25 00 01	System Control and Monitoring	For the lump sum of		$
3-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
3-9	26 05 33	Electrical Conduit	200 ft.	$	$
3-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
3-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
3-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
3-13	27 30 01	Telephone System	For the lump sum of		$
3-14	28 10 01	Security and Surveillance System	For the lump sum of		$
3-15	31 02 10	Water for Dust Abatement	19 M (1,000) Gal	$	$
3-16	31 02 30	Dust Palliative	0.25 acre	$	$
3-18	31 37 00	Riprap	125 yd^3	$	$
3-19	32 15 10	Gravel Surfacing	135 yd^3	$	$

PRICE SCHEDULE 3 (Naschitti)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
3-20	32 31 19	Steel Ornamental Fence	244 ft.	$	$
3-21	32 92 20	Seeding	0.03 acre	$	$
~~***3-22***~~	~~***33 11 12***~~	***Deleted***			
3-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
3-24	35 21 95	Steel Manifold Piping	For the lump sum of		$
3-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 3 $_____

PRICE SCHEDULE 4 (BBN9)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
4-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
4-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
4-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
4-4	01 78 30	Final Drawings	For the lump sum of		$5,000
4-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
4-6	13 34 00	Turnout Buildings	For the lump sum of		$
4-7	25 00 01	System Control and Monitoring	For the lump sum of		$
4-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
4-9	26 05 33	Electrical Conduit	220 ft.	$	$

PRICE SCHEDULE 4 (BBN9)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
4-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
4-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
4-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
4-13	27 30 01	Telephone System	For the lump sum of		$
4-14	28 10 01	Security and Surveillance System	For the lump sum of		$
4-15	31 02 10	Water for Dust Abatement	19 M (1,000) Gal	$	$
4-16	31 02 30	Dust Palliative	0.25 acre	$	$
4-17	31 23 50	Low Water Crossing	For the lump sum of		$
4-19	31 37 00	Riprap	52 yd^3	$	$
4-20	32 15 10	Gravel Surfacing	81 yd^3	$	$
4-21	32 31 19	Steel Ornamental Fence	204ft.	$	$
4-22	32 92 20	Seeding	0.13 acre	$	$
4-23	33 11 12	Line Pipe for BBN9 Turnout	For the lump sum of		$
4-24	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
4-25	35 21 95	Steel Manifold Piping	For the lump sum of		$
4-26	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 4 $_____

PRICE SCHEDULE 5 (Coyote Canyon)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
5-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
5-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
5-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
5-4	01 78 30	Final Drawings	For the lump sum of		$5,000
5-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
5-6	13 34 00	Turnout Buildings	For the lump sum of		$
5-7	25 00 01	System Control and Monitoring	For the lump sum of		$
5-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
5-9	26 05 33	Electrical Conduit	310 ft.	$	$
5-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
5-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
5-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
5-13	27 30 01	Telephone System	For the lump sum of		$
5-14	28 10 01	Security and Surveillance System	For the lump sum of		$
5-15	31 02 10	Water for Dust Abatement	12 M (1,000) Gal	$	$
5-16	31 02 30	Dust Palliative	0.16 acre	$	$
5-17	31 23 50	Access Road to Coyote Canyon	For the lump sum of		$
5-19	31 37 00	Riprap	53 yd^3	$	$

PRICE SCHEDULE 5 (Coyote Canyon)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
5-20	32 15 10	Gravel Surfacing	67 yd^3	$	$
5-21	32 31 19	Steel Ornamental Fence	189 ft.	$	$
5-22	32 92 20	Seeding	0.05 acre	$	$
5-23	33 11 12	Line Pipe for Coyote Canyon Turnout	For the lump sum of		$
5-24	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
5-25	35 21 95	Steel Manifold Piping	For the lump sum of		$
5-26	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 5 $_____

PRICE SCHEDULE 6 (Rock Springs)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
6-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
6-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
6-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
6-4	01 78 30	Final Drawings	For the lump sum of		$5,000
6-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
6-6	13 34 00	Turnout Buildings	For the lump sum of		$
6-7	25 00 01	System Control and Monitoring	For the lump sum of		$

PRICE SCHEDULE 6 (Rock Springs)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
6-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
6-9	26 05 33	Electrical Conduit	370 ft.	$	$
6-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
6-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
6-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
6-13	27 30 01	Telephone System	For the lump sum of		$
6-14	28 10 01	Security and Surveillance System	For the lump sum of		$
6-15	31 02 10	Water for Dust Abatement	13.5 M (1,000) Gal	$	$
6-16	31 02 30	Dust Palliative	0.18 acre	$	$
6-18	31 37 00	Riprap	58 yd^3	$	$
6-19	32 15 10	Gravel Surfacing	97 yd^3	$	$
6-20	32 31 19	Steel Ornamental Fence	199 ft.	$	$
6-21	32 92 20	Seeding	0.03 acre	$	$
6-22	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
6-23	35 21 95	Steel Manifold Piping	For the lump sum of		$
6-24	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 6 $_____

PRICE SCHEDULE 7 (Window Rock)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
7-1	WBR 1452.232-81	Mobilization and Preparatory Work	For the lump sum of		$
7-2	01 56 15	Utility Crossing Investigations	For the lump sum of		$
7-3	01 78 30	As-built Drawings	For the lump sum of		$10,000
7-4	01 78 30	Final Drawings	For the lump sum of		$5,000
7-5	***01 80 15***	***Commissioning***	***For the lump sum of***		***$***
7-6	13 34 00	Turnout Buildings	For the lump sum of		$
7-7	25 00 01	System Control and Monitoring	For the lump sum of		$
7-8	26 05 02	Complete Electrical System for Outdoor Equipment Only	For the lump sum of		$
7-9	26 05 33	Electrical Conduit	370 ft.	$	$
7-10	26 05 33	Fabricated Sheet Steel Boxes, Wiring Troughs, and Precast Handholes	For the lump sum of		$
7-11	26 42 00	Corrosion System(s) for Turnout Yard Pipe	For the lump sum of		$
7-12	26 42 11	Buried Galvanic Anode Cathodic Protection for Turnout Yard Pipe	For the lump sum of		$
7-13	27 30 01	Telephone System	For the lump sum of		$
7-14	28 10 01	Security and Surveillance System	For the lump sum of		$
7-15	31 02 10	Water for Dust Abatement	10 M (1,000) Gal	$	$
7-16	31 02 30	Dust Palliative	0.14 acre	$	$
7-17	31 23 50	Access Road to Window Rock	For the lump sum of		$
7-19	31 37 00	Riprap	64 yd^3	$	$

PRICE SCHEDULE 7 (Window Rock)

CLIN	Section	Supplies or Services	Quantity and Unit	Unit Price	Amount
7-20	32 15 10	Gravel Surfacing	59 yd^3	$	$
7-21	32 31 19	Steel Ornamental Fence	165 ft.	$	$
7-22	32 92 20	Seeding	0.03 acre	$	$
7-23	33 13 00	Disinfection of Water Utility Distribution	For the lump sum of		$
7-24	35 21 95	Steel Manifold Piping	For the lump sum of		$
7-25	35 22 14	Valves and Equipment	For the lump sum of		$

TOTAL FOR PRICE SCHEDULE 7 $_____

END OF SUPPLIES OR SERVICES AND PRICES

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PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 WORK RESTRICTIONS AND LIMITATIONS

A. Allowable Work Hours: Sunrise to sunset, unless otherwise authorized by COR.

B. Comply with noise and light restrictions listed in Section 01 57 20 – Environmental Controls and 01 35 20 – Safety and Health.

C. Perform work in roadways in accordance with Section 01 55 20 – Traffic Control.

D. Complete work on San Juan Lateral by January 1, 2028, to accommodate flushing and disinfection of the lateral.

1. Install tee and buried valve at Coyote Canyon.
2. Install valves at BBN9. Refer to Drawing 1695-D-62548.
3. Replace blind flanges on existing turnouts with valves.
4. Limit outage to San Juan Lateral to no more than 28 days. Coordinate outage with COR. Notify COR 28 days before outage.
    a. It is anticipated potable water will be available for delivery to CWPS on or about

E. San Juan Lateral Water Treatment Plant will begin delivering potable water in late 2028.

1. NTUA will flush and disinfect approximately two existing Consecutive Public Water System(s)(CWPS) each month.

2. Coordinate commissioning for each turnout with COR. Water Treatment Plant Contractor and NTUA. Commission a maximum of two turnouts per month, beginning January 1, 2029 and finishing in April, 2029.

F. Complete connection to lateral and (CWPS), including buildings, and equipment for BBN9, Coyote Canyon, and Naschitti by September 1, 2028. Except:

1. Work with WTP contractor to install Government Furnished Equipment NIP by April 31, 2028. Notify COR 28-days prior when building is on-site and ready for panel installation.

2. WTP contractor will support completing communications link (Lateral Integration Test) to the San Juan Lateral SCADA system by August 30, 2028, but not earlier than August 1, 2028.

a. Turnout Contractor shall provide onsite support for the WTP Contractor in communication link work.

b. COR will notify Turnout Contractor 28 days prior to WTP Contractor arrival on-site.

c. Refer to Section 25 00 01 – System Control and Monitoring for NIP installation coordination and requirements and 01 80 15 – Commissioning for Lateral Integration Testing requirements.

3. Complete commissioning of turnout facilities and water delivery to downstream CPWS by February 28, 2029.

G. Complete connection to lateral and CWPS, including buildings, and equipment for remaining facilities by March 15, 2029.

1. Work with WTP contractor to install Government Furnished Equipment NIP by January 15, 2029. Notify COR 28-days prior when building is on-site and ready for panel installation.

2. WTP contractor will support completing communications link (Lateral Integration Test) to the San Juan Lateral SCADA system by March 30, but not earlier than March 1, 2029.

a. Turnout Contractor shall provide onsite support for the WTP Contractor in communication link work.

b. COR will notify Turnout Contractor 28 days prior to WTP Contractor arrival on-site.

c. Refer to Section 25 00 01 – System Control and Monitoring for NIP installation coordination and requirements and 01 80 15 - Commissioning for Lateral Integration Testing requirements.

3. Complete commissioning of turnout facilities and water delivery to downstream CPWS by April 15, 2029.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 31 19
PROJECT MANAGEMENT AND COORDINATION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (Reclamation)

1. RSHS — Reclamation Safety and Health Standards, including revisions, available at https://www.usbr.gov/safety/rshs/index.html

1.03 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals.

B. RSN 01 31 19-1, Written Summary:

1. Written weekly summary of upcoming work, traffic plans and progress meeting minutes. Provide work schedule for next 3 weeks including list of RSNs and RFIs.

C. RSN 01 31 19-2, Off ROW Land Use Coordination:

1. Show expanded use location and extent of impact.
2. Request cultural approvals:
   a. In accordance with Section 01 57 90 – Preservation of Historical and Archeological Data.
   b. Minimum information required for cultural analysis: Map and UTM data in NAD83 format of the location, narrative of intended use of the new area, summary of intended access to the new area, and a need by date.
3. Request or revise Environmental Permits in accordance with:
   a. Section 01 57 20 – Environmental Controls.
   b. Section 01 57 30 – Water Pollution Control.
   c. Section 01 57 50 – Tree and Plant Protection.
   d. Section 01 57 60 – Protected Species.
4. Do not use proposed land until approved by COR:
   a. Approvals may require up to 6 months.

5. Requirement does not apply to existing commercial facilities.

D. RSN 01 31 19-3, Key Personnel:

1. Submit list of key personnel:
   a. Project Manager.
   b. Site Superintendent.
   c. Contractor's Onsite Safety Representative.
   d. Quality Control Supervisor.
2. Include resume of Project Manager and Site Superintendent.
3. CO approval of key personnel required prior to work being performed.
4. Personally Identifiable Information (PII) shall be encrypted using WINZip or Adobe Acrobat before transmitting electronically.

E. RSN 01 31 19-4, Replacement of Key Personnel:

1. Submit key personnel changes with a minimum notice of 48 hours to CO.
2. CO approval of changes in key personnel required prior to work being performed by new personnel.

1.04 QUALIFICATIONS

A. Project Manager

1. Comparable role on three projects of similar nature, size, and complexity.
2. Employed by prime contractor.

B. Site Superintendent:

1. Comparable role on three projects of similar nature, size, and complexity.
2. Employed by prime contractor.

C. Contractor's Onsite Safety Representative: Meet requirements of Section 01 35 30 – Contractor's Onsite Safety Personnel.

D. Quality Control Supervisor: Meet requirements of Section 01 46 00 – Quality Procedures.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROJECT COORDINATION

A. Coordinate work with COR, other Bureau of Reclamation (Reclamation) staff, Bureau of Indian Affairs, and other Federal, Tribal, State, and local agencies, as needed.

B. Facilitate onsite work with above listed agencies.

C. 48-Hour Look Ahead Notification:

1. Provide COR, OGR, Cultural Contract COR, and Cultural Resource Contractor via email, and phone with a 48-hour notice when Cultural Resource Contractor monitoring services are required.
2. Contact information including Cultural Contract COR and Cultural Resource Contractor will be provided at preconstruction meeting.

3.02 MEETINGS

A. The following meetings are considered significant, but do not relieve Contractor from responsibility of other meetings required by contract:

1. Preconstruction Meeting:
    a. Meet with Government prior to start of construction. Meeting to occur after approval of RSN 01 35 20-1, Safety Program.
    b. Provide, as a minimum, Project Manager, Project Superintendent, and Project Engineer for Preconstruction Meeting.
    c. Review will include, but may not be limited to:
        1) Use of premises by Contractor, Government, and public:
            a) Off ROW land use coordination.
        2) Construction facilities and controls.
        3) Access to work and haul routes.
        4) Temporary utilities.
        5) Survey layout and installation of temporary fencing of the entire project work area prior to surveying.
        6) Security.
        7) Housekeeping procedures.
        8) Schedules and sequence of work.
        9) Procedures for testing.
        10) Procedures for maintaining record documents.
        11) Special site requirements:

a) Archeological: Refer to Section 01 57 90 – Preservation of Historical and Archaeological Data.

b) Biological: Refer to Section 01 57 60 – Protected Species.

c) Environmental: Refer to:

i. Section 01 57 20 – Environmental Controls

ii. Section 01 57 30 – Water Pollution Control.

12) Roles and responsibilities:

a) Government organization and personnel.

b) Contractor organization and personnel.

c) Subcontractors.

d) Proposed work schedule.

13) Contract requirements:

a) Progress payments.

b) Invoices.

c) Differing site conditions.

d) Changes.

e) Superintendence by Contractor.

f) Payment for mobilization and preparatory work.

g) Submittal procedures.

h) Emails and telephone calls.

2. Cultural Resource Sensitivity Awareness Training:

a. Refer to Section 01 57 90 – Preservation of Historical and Archaeological Data.

b. Complete Government-provided training before initial groundbreaking. Schedule for training will be mutually agreed upon.

3. Preconstruction Safety Meeting:

a. Meet with COR prior to start of construction. Meeting to occur after approval of RSN 01 35 20-1, Safety Program.

b. Review will include:

1) Safety requirements.

2) Protection of cultural resources and cultural sensitivity training requirements.

3) Monthly joint safety policy meetings.

4) Weekly toolbox safety meetings.

5) Safety program.

6) Job hazard analysis.

7) Required safety inspections.

8) Government will monitor Contractor's operations for compliance with RSHS.

4. Progress Meetings:

a. Meet weekly with Government to review:

1) Progress made.

2) Review as-built drawings.

3) Difficulties in performing work.

4) Safety performance.

5) Resolution necessary to meet specifications requirements.

6) Status of contract documents (e.g., RFI's, submittals).

7) Three-week look ahead, including installing temporary safety fencing and upcoming cultural monitoring areas discussed.

8) Updating of progress reports.

9) Other items as determined by COR and OGR.

b. Develop meeting agenda and minutes, may be used as Written Summary.

5. SJWTP Coordination Meetings:

a. Meet monthly with Government, SJWTP Contractor, and NTUA to discuss coordination issues such as SCADA Integration and Commissioning.

b. Develop meeting agenda and minutes, may be used as Written Summary.

END OF SECTION

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SECTION 01 33 00
SUBMITTALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME Y14.1-2020	Decimal Inch Drawing Sheet Size and Format

B. National Institute of Building Sciences (NIBS)

1.	NIBS NCS-2014	United States National CAD Standards, Version 6

1.03 DEFINITIONS

A. Days: Calendar days.

B. Required Submittal Number (RSN): Identifies items to be submitted together as a complete submittal.

C. Submittal Types, as listed in Table 01 33 00A – List of Submittals:

1. A – Action:
    a. Government will respond as to adequacy of submittal.
    b. Action submittals: Considered to be "shop drawings" within terms of the clause at FAR 52.236-21, Specifications and Drawings for Construction.
2. I – Informational:
    a. Government will acknowledge receipt of Informational submittals.
    b. Government may reject an Informational submittal when submittal does not comply with contract. Contractor shall correct mistakes or deficiencies in rejected Informational submittals and resubmit.
    c. Informational submittals: Considered to be "shop drawings" within terms of the clause at FAR 52.236-21, Specifications and Drawings for Construction, except that approval by Government is not required.

1.04 SUBMITTAL REQUIREMENTS

A. In case of conflict between requirements of this section and requirements included elsewhere in these specifications, requirements included elsewhere take precedence.

B. General:

1. Prepare in English.
2. Label with contract number, title, and RSN.
3. Measurement units: US Customary Units.
4. Provide unique transmittal number for each submittal.

C. Drawings:

1. Minimum identification in title block:
   a. Contract number and title.
   b. Contractor's or supplier's title and drawing number.
   c. Date.
2. Reserve 3- by 3-inch space next to title block for review stamps.
3. Print Size: D size (22- by 34-inches) as defined by ASME Y14.1.
4. Draw to scale using computer drafting or drafting equipment, unless otherwise specified:
   a. Computer drafted drawings:
      1) In accordance with NIBS NCS.
      2) Electronic file format: Compatible with AutoCAD, Version 2018 or newer.
      3) Compile using "eTransmit" utility in AutoCAD.
   b. Drawing prepared with drafting equipment, when allowed: Lettering shall be neat.
5. Drawings designated as "Government Format" in specifications:
   a. Computer drafted.
   b. Government will provide electronic AutoCAD format template.
   c. Title block and sheet format:
      1) As shown on Standard Drawing 40-D-7102.
      2) Government will provide template.
      3) Government will provide specific title block information to be used.

6. Final Drawings:

   a. Computer drafted.

   b. Government will provide electronic AutoCAD format template.

   c. Show as-built changes, including revision dates, made during installation. Indicate changes by clouding.

   d. Electronic Files: ~~On~~ Reclamation **issued CDMS. ~~issued USB Drive.~~**

D. Product Data:

1. Mark manufacturer's data for commercial products or equipment, such as catalog cut sheets:

   a. Identify manufacturer's name, type, model, size, and characteristics.

   b. Illustrate that product or equipment meets requirements of specifications.

   c. Mark items to be furnished in a manner that will photocopy (no highlighter).

   d. Strike through items that do not apply.

E. Certifications:

1. Certifications by a registered professional: Signed and sealed by registered professional:

   a. Not required for non-design work.

2. Manufacturer’s certifications: Signed by authorized representative of manufacturer.

F. Manuals:

1. Copies:

   a. Printed copies: Bound and indexed.

   b. Electronic copies: Searchable Adobe .pdf on Reclamation issued **CDMS. ~~USB Drive~~**:

      1) Bookmark longer files to assist in navigating file.

      2) Electronic files may be submitted as separate parts of a manual. After each separate electronic file is approved, compile each approved electronic file to assemble an entire manual which shall be consistent with final manual submittal.

2. Contents:

   a. Parts identification lists, lists of special tools, and accessories.

   b. Schematics and wiring diagrams.

   c. Detailed instructions for installing, operating, lubricating, and maintaining equipment.

d. As-built drawings, photographs, and test records or reports if required by specifications.

G. Photographs:

1. Digital files: In .jpeg or similar format.

1.05 SUBMITTALS PROCEDURES

A. Submit only checked submittals. Submittals without evidence of Contractor's approval will be returned for resubmission.

B. Submit complete sets of required materials for each RSN as specified in "Submittals Required" column in Table 01 33 00A – List of Submittals. A complete set includes all listed items for RSNs with multiple parts.

C. Submit sets specified in "Sets to be sent:" columns in Table 01 33 00A – List of Submittals:

1. Submit official corresponding and construction documentation via Government provided Construction Document Management System. Documents include, but are not limited to:

a. Submittals.

b. RFCs : Request for Change (Construction Contractor generated document).

c. RFIs: Request for Information (Construction Contractor generated document).

d. RFPs: Request for Proposal (Government generated document).

e. CMs: Contract Modifications.

f. Construction Contractor Proposals.

g. Value Engineering Proposals.

h. Invoices and Progress Payments.

i. Construction Contract Schedules.

j. Daily Inspection Reports

k. Meeting Agendas and Meeting Minutes.

l. Letters and Memos.

2. Deleted

3. Submittals identified with "CDMS" in the top row of the "Sets to be sent" column require approval (Action) or acknowledgement (Informational) before sending required sets of hardcopy submittals.

4. Submittals identified with "CDMS" in the bottom row of the "Sets to be sent" column require submittal of hardcopy submittals before finalizing submittal in CDMS.

D. Include the following information in transmittal letters:

1. Contract number and title.
2. RSN for each attached submittal.
3. Responsible code.
4. Number of sets for each RSN.
5. Identify submittal as initial or resubmittal.

E. Resubmittal of submittals not approved:

1. Mark changes such that they are readily identifiable and show revision date.
2. Describe reasons for significant changes in transmittal letter.
3. Resubmit returned submittals within 28 days after receiving the comments, unless otherwise directed.
4. Requirements for initial submittals apply to resubmittals.

1.06 REVIEW OF SUBMITTALS

A. Time Required:

1. Submittal review will require 28 days for review of each submittal or resubmittal, unless otherwise specified.
2. Time required for review of each submittal or resubmittal begins when complete sets of materials required for a particular RSN are received and extends through digital signature date stamp and upload date in CDMS.

B. Time in Excess of Specified:

1. CO may extend contract completion date to allow additional time for completing work affected by excess review time:
   a. Time extension will be to extent that excess review time caused delay to contract completion date.
   b. Time extension will not exceed time used in excess of specified number of days for review of submittals or resubmittals.
   c. Concurrent days of excess review time resulting from review of two or more separate submittals or resubmittals will be counted only once in extending contract completion date.
2. No time extension will be allowed if Contractor fails to make complete action submittals in sequence and within time periods specified.
3. Adjustment for delay will be made only to extent that:

a. Approval was required under contract, and

b. Requests for approval were properly and timely submitted and were approved.

4. Adjustment will be subject to terms of paragraphs (b) and (c) of the clause at FAR 52.242-14, Suspension of Work; however, no such delay shall be deemed to be a "suspension order" as the term is used in that clause.

C. Return of Submittals:

1. Return of submittals will be by CDMS response or hard copy, as applicable.

2. Action Submittals: One set of submittals required for action will be returned either approved, partially approved, approved subject to identified changes and resubmit, or not approved:

a. Submittals that are not approved shall be revised and resubmitted.

b. Do not change designs without approval of CO after drawings, documentation, and technical data have been approved.

c. Exception: Schedules will be either accepted, partially accepted or not accepted.

3. Informational Submittals: Government will acknowledge Informational submittals:

a. Informational submittals will not be returned when they comply with specifications.

b. Informational submittals that do not comply with specifications may be returned for resubmittal or additional information may be requested.

1.07 HARD COPY OF TRANSMITTALS

A. Addresses for codes listed in Table 01 33 00A – List of Submittals:

1. Contracting Officer, Bureau of Reclamation, Attn: UC-840, 125 South State St., Room 6107, Salt Lake City, UT 84138-1147.

2. Construction Engineer, Bureau of Reclamation, 1235 La Plata Hwy, Farmington, New Mexico, 87401.

3. Technical Service Center, Bureau of Reclamation, Attn: 86-68160, PO Box 25007, Denver CO 80225-0007; Express Mail: Sixth and Kipling, Building 67, Room 152.

4. Navajo Tribal Utility Authority (NTUA), Attn: Operations Manager, PO Box 1749, Shiprock, NM 87420.

B. Send original transmittal letter with appropriate number of sets to office listed in "Responsible Code" column in Table 01 33 00A – List of Submittals:

1. Responsible codes starting with "86-6" are located in the Technical Service Center. Send these submittals to the TSC address shown above.

C. Send copy of transmittal letter with appropriate number of sets to offices that are not the responsible code but show "No. of sets to be sent to" in Table 01 33 00A – List of Submittals.

D. Provide copy of transmittal letter on CDMS.

E. When "No. of sets to be sent to" is 0, send a copy of transmittal letter to that office.

F. Submittals required by the specifications, but not listed in Table 01 33 00A – List of Submittals:

1. Submit in accordance with this section.
2. Submit to CE, unless otherwise specified.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 14 10-1	Use of Site	Land Use and Landscape Rehabilitation Plan	At least 63 days before use of ROW	A	CE	CDMS			
01 14 10-2	Use of Site	Optional Onsite Security Plan	At least 60 days before use of site	A	CE	CDMS			
01 31 19-1	Project Management and Coordination	Written Summary	Monthly once onsite construction begins	A	CE	CDMS			
01 31 19-2	Project Management and Coordination	Off ROW Land Use Coordination	At least 6 months before use of OFF ROW land	I	CE	CDMS			
01 31 19-3	Project Management and Coordination	Key Personnel	Within 7 days after receipt of Notice of Award	A	CE	CDMS			
01 31 19-4	Project Management and Coordination	Replacement of Key Personnel	A minimum of 2 business days prior to work being performed by new personnel	A	CE	CDMS			
~~***01 31 30-1***~~	***Deleted***								
~~***01 31 30-2***~~	***Deleted***								
01 32 10-1	Construction Program	Representative Information	Within 7 days after receipt of Notice of Award	I	CE	CDMS			
01 32 10-2	Construction Program	Baseline Schedule	Within 21 days after receipt of initial Notice to Proceed	A	CE	CDMS			
01 32 10-3	Construction Program	Updated Schedule Reports	Monthly or with Progress Payments	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 32 10-4	Construction Program	Time Impact Analysis	Within 28 days after CO directs a contract change, with any proposal for a future modification, with any value engineering proposal, or with any request or claim for an equitable adjustment to the contract.	A	CE	CDMS			
01 35 10-1	Safety Data Sheets	Complete LHM and SDS	At least 14 days before jobsite delivery of hazardous material	I	CE	CDMS			
01 35 10-2	Safety Data Sheets	Updated LHM and SDS	At least 14 days before jobsite delivery of hazardous material not previously listed	I	CE	CDMS			
01 35 20-1	Safety and Health (WBR 1452.223-81)	Safety Program	Submitted and accepted before commencing onsite work. Submit at least 45 days before beginning onsite work. See Section 3 of RSHS	A	CE	CDMS			
01 35 20-2	Safety and Health	Job Hazard Analyses	At least 14 days before commencing onsite activities for each phase of work	A	CE	CDMS			
01 35 20-3	Safety and Health	Monthly Accident Summary Report	First day of each month. See paragraph 3.8 of RSHS	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 35 30-1	Contractor's Onsite Safety Personnel	Resumes	At least 45 days before beginning onsite construction; resubmit upon personnel change	A	CE	CDMS			
01 35 30-2	Contractor's Onsite Safety Personnel	Safety Inspection Reports	At least once each week	I	CE	CDMS			
01 46 00-1	Quality Procedures	Definable Features of Work List	Within 21 days after receipt of initial Notice to Proceed	A	CE	CDMS			
01 46 00-2	Quality Procedures	Contractor Quality Control (QC) Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 46 00-3	Quality Procedures	Contractor's Quality Daily Report	Weekly	I	CE	CDMS			
01 46 00-4	Quality Procedures	Quality Control Supervisor (QCS) Resume	At least 45 days before beginning onsite work. Resubmit upon personnel change.	A	CE	CDMS			
01 46 00-5	Quality Procedures	Contractor's Quality Testing Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 46 00-6	Quality Procedures	Summary of Monthly Test Results	Monthly, on or before 5th day of each month	A	CE	CDMS			
01 46 20-1	Testing Agency Services	Testing Agencies Services Plan	At least 45 days before testing is required	A	CE	CDMS			
01 46 25-1	Special Inspections	Preconstruction Submittals	At least 45 days before special inspection is required	A	CE	CDMS			
01 46 25-2	Special Inspections	Test Reports	Daily and Biweekly	A	CE	CDMS			
01 46 25-3	Special Inspections	Certificates	At least 45 days before special inspection is required	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 46 25-4	Special Inspections	Closeout Submittals	Within 14 days of completion of work	A	CE	CDMS			
01 55 00-1	Vehicular Access and Parking	Pre-Construction Digital Recording	At least 21 days before beginning onsite work	A	CE	0	1	0	0
01 55 00-2	Vehicular Access and Parking	Right of Way (ROW) Access Plan	At least 21 days before beginning onsite work.	A	CE	CDMS			
01 55 00-3	Vehicular Access and Parking	Post-Construction and Post-Repair Digital Recording	At least 21 days before leaving site.	A	CE	0	1	0	0
01 55 20-1	Traffic Control	Information of Traffic Control Plan (TCP)	At least 70 days before affecting public traffic.	A	CE	CDMS			
01 55 20-2	Traffic Control	Approved NMDOT Permit(s)	At least 45 days before affecting public traffic.	A	CE	CDMS			
01 56 10-1	Protection of Existing Installations	Plan for Protecting Existing Installations	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-1	Protection of Existing Utilities	Utility Owner Acknowledgement	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-2	Protection of Existing Utilities	Utility Crossing Investigation	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 56 15-3	Protection of Existing Utilities	Work Plan within Utility Easements	At least 45 days before start of onsite construction work	A	CE	CDMS			
01 57 20-1	Environmental Controls	Copy of Applicable Air Quality Permit	At least 14 days before beginning onsite work	I	CE	CDMS			
01 57 30-1	Water Pollution Control	Stormwater Pollution Prevention Plan (SWPPP)	At least 28 days before start of onsite construction work	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 57 30-2	Water Pollution Control	Spill Prevention, Control, and Countermeasure (SPCC) Plan	At least 28 days before delivery or storage of oil	A	CE	CDMS			
01 57 40-1	Pesticides	Qualifications	At least 45 days before beginning onsite work	A	CE	CDMS			
01 57 40-2	Pesticides	Pesticide Use Plan	At least 45 days before application of first pesticide	A	CE	CDMS			
01 57 50-1	Tree and Plant Protection	Protection Plan	At least 45 days before beginning onsite work	A	CE	CDMS			
01 57 90-1	Preservation of Historical and Archeological Data	Alternate Use Area	At least 45 days before beginning offsite work, if commercial site; if non-commercial site then at least 6-months prior to use of land	A	CE	CDMS			
01 61 10-1	Buy American	Buy American Certificates	Prior to project closeout	I	CE	CDMS			
01 71 20-1	Surveying	Surveying Plan	Within 45 days after Notice to Proceed, and approved before beginning survey work	A	CE	CDMS			
01 71 20-2	Surveying	Resume(s)	At least 45 days before beginning survey work; resubmit upon personnel change	I	CE	CDMS			
01 71 20-3	Surveying	Accuracy Check Results	At least 45 days before beginning survey work	I	CE	CDMS			
01 74 00-1	Cleaning and Waste Management	Waste Production and Disposal Plan	At least 45 days before beginning onsite work	I	CE	CDMS			

Table 01 33 00A – List of Submittals

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
01 74 00-2	Cleaning and Waste Management	Waste Production and Disposal Records	Within 7 days of hazardous waste disposal	I	CE	CDMS			
01 74 00-3	Cleaning and Waste Management	Environmental Consultant Resume	At least 45 days before beginning environmental assessment	I	CE	CDMS			
01 74 00-4	Cleaning and Waste Management	Environmental Site Assessment	Within 14 days of completion of work	I	CE	CDMS			
01 78 30-1	Project Record Documents	Electronic Media Release Form, Upon Request	With request for Electronic Drawings	A	CE	CDMS			
01 78 30-2	Project Record Documents	Progress As-Built Drawings	Monthly once onsite construction begins.	A	CE	CDMS			
01 78 30-3	Project Record Documents	Final As-Built Drawings	Within 28 days of Government issued substantial completion of work	A	CE	CDMS			
01 78 30-4	Project Record Documents	Final Drawings	Within 14 days of approval of RSN 01 78 30-3	A	CE	0	2	0	2
						CDMS			
01 80 15 -1	Commissioning	Qualifications	At least 60 days before start of commissioning	A	CE	CDMS			
01 80 15 -2	Commissioning	Testing and Commissioning Plan	At least 60 days before start of commissioning	A	CE	CDMS			
01 80 15 -3	Commissioning	Testing and Commissioning Reports	Within 14 days of completion of Testing	I	CE	CDMS			
01 80 15 -4	Commissioning	Post-Testing and Commissioning Documentation	Within 28 Days of completion of Testing	I	CE	CDMS			
03 11 10-1	Concrete Forming	Formwork Design and Drawings	At least 45 days prior to installation work	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
03 20 00-1	Concrete Reinforcement	Reinforcement Diagrams and Lists	At least 45 days before placing concrete	A	CE	CDMS			
03 20 60-1	Concrete Anchors	Instructions	At least 45 days before placing concrete	A	CE	CDMS			
03 20 60-2	Concrete Anchors	Test Data and Results	At least 7 days before testing	A	86-8530	CDMS			
03 30 00-1	Cast-In-Place Concrete	Approval Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-2	Cast-In-Place Concrete	Concrete Placement Schedule	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-3	Cast-In-Place Concrete	Certifications	At least 45 days before placing concrete	A	86-8530	CDMS			
03 30 00-4	Cast-In-Place Concrete	Test Reports	At least 45 days before placing curing compound	A	86-8530	CDMS			
03 48 00-1	Precast Concrete Manholes	Certificates	At least 45 days before placing concrete	A	86-8530	CDMS			
03 48 00-2	Precast Concrete Manholes	Shop Drawings	At least 45 days before placing concrete	A	86-8530	CDMS			
03 48 00-3	Precast Concrete Manholes	Product Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 62 20-1	Nonshrink Grout for Equipment and Metalwork	Packaged Nonshrink Grout Data	At least 45 days before placing concrete	A	86-8530	CDMS			
03 63 00-1	Epoxy Grout	Approval Data	At least 45 days before beginning grouting	A	86-8530	CDMS			
09 96 20-1	Coatings	Approval Data	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-2	Coatings	Final Approval Data	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-3	Coatings	Paint Chip Samples	At least 28 days before beginning	A	86-8540	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

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RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
			coating work			0	1	1	0
09 96 20-4	Coatings	Qualifications	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-5	Coatings	Quality Control Plan	At least 28 days before beginning coating work	A	86-8540	CDMS			
09 96 20-6	Coatings	Contractor Quality Testing Report	At least 14 days after completion of test	A	86-8540	CDMS			
10 14 00-1	Signage	Shop Drawings and Approval Data	At least 28 days before procurement	A	86-8120	CDMS			
10 14 26-1	Utility Markers	Utility Marker Plan	At least 28 days before procurement	A	CE	CDMS			
10 44 20-1	Fire Extinguishers	Approval Data	At least 28 days before procurement	A	CE	CDMS			
13 34 00-1	Prefabricated Engineered Structures	Approval Drawings and Data	At least 28 days before procurement	A	CE	CDMS			
13 34 00-2	Prefabricated Engineered Structures	Sample and Color Sample Kit	At least 28 days before procurement	A	CE	CDMS			
						0	1	1	0
13 34 00-3	Prefabricated Engineered Structures	Geotechnical Report	At least 28 days before procurement	A	86-68314	CDMS			
13 34 00-4	Prefabricated Engineered Structures	Factory Test Reports	Within 14 days after completion of test	A	86-68460	CDMS			
13 34 00-5	Prefabricated Engineered Structures	Field Quality Test Reports	Within 14 days after completion of test	A	86-68460	CDMS			
13 34 00-6	Prefabricated Engineered Structures	Final Data	At least 28 days before procurement	A	86-68460	CDMS			
25 00 01-1	System Control and Monitoring	Qualifications	At least 28 days before beginning work	A	CE	CDMS			

Table 01 33 00A – List of Submittals

* Submittal types: A – Action, I – Information

** CO indicates Contracting Officer, CE indicates Construction Engineer, TSC indicates Technical Service Center, and NTUA indicates the City of Gallup New Mexico.

RSN	Clause or Section Title	Submittals Required	Due Date or Delivery Time	Type *	Respon-sible Code	No. of sets to be sent to:**			
						CO	CE	TSC	NTUA
25 00 01-2	System Control and Monitoring	System Description Documentation	At the same time as equipment is shipped to the site	A	CE	CDMS			
25 00 01-3	System Control and Monitoring	Approval Drawings and Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
25 00 01-4	System Control and Monitoring	Layout Drawings	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
25 00 01-5	System Control and Monitoring	Software Development Documentation	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	I	CE	CDMS			
25 00 01-6	System Control and Monitoring	Final Drawings, Documentation, and Operations and Maintenance Manuals	Within 14 days of completion of work	I	CE	CDMS			
~~***25 08 10-1***~~	***Deleted***								
~~***25 08 10-2***~~	***Deleted***								
~~***25 08 10-3***~~	***Deleted***								
~~***25 08 10-4***~~	***Deleted***								
~~***25 08 10-5***~~	***Deleted***								
26 05 10-1	Conductors and Cables	Approval Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 05 10-2	Conductors and Cables	Field Test Reports	No later than 14 days after completing testing	A	86-68460	CDMS			
26 05 20-1	Grounding and Bonding	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			

Table 01 33 00A – List of Submittals

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						CO	CE	TSC	NTUA
26 05 20-2	Grounding and Bonding	As-built Drawings	No later than 28 days after completing work	A	86-68460	CDMS			
26 05 20-3	Grounding and Bonding	Test Reports	No later than 14 days after completing testing	A	86-68460	CDMS			
26 05 33-1	Raceways and Boxes	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 05 33-2	Raceways and Boxes	As-built Drawings	No later than 28 days after completing work	A	86-68460	CDMS			
26 27 10-1	Meter Socket	Approval Drawings and Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 28 20-1	Service Disconnect Switch	Approval Data	Submitted and approved at least 28 days before procurement	A	86-68460	CDMS			
26 42 00-1	Cathodic Protection Systems	Certification and Data	At least 42 days before procuring materials	A	86-68540	CDMS			
26 42 11-1	Buried Galvanic Anode Cathodic Protection	Certification and Data	At least 42 days before procuring materials	A	86-68540	CDMS			
26 42 11-2	Buried Galvanic Anode Cathodic Protection	Final Data	Within 28 days of final testing	A	86-68540	CDMS			
27 30 01-1	Telephone System	Communication System Design	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
27 30 01-2	Telephone System	Approval Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			

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						CO	CE	TSC	NTUA
27 30 01-3	Telephone System	Test Procedure and Test Schedule	At least 28 days before beginning test	I	CE	CDMS			
27 30 01-4	Telephone System	Operation and Maintenance Manual	At least 28 days before beginning test	I	CE	CDMS			
27 30 01-5	Telephone System	Field Operational Checkout Test Report	At least 14 days after completion of test	A	CE	CDMS			
28 10 01-1	Security and Surveillance System	Approval Drawings and Data	Within 150 days after Notice to Proceed, and at least 45 days prior to installation	A	CE	CDMS			
28 10 01-2	Security and Surveillance System	Test Procedure and Test Schedule	At least 28 days before beginning test	I	CE	CDMS			
28 10 01-3	Security and Surveillance System	Test Report	At least 14 days after completion of test	A	CE	CDMS			
28 10 01-4	Security and Surveillance System	Final Drawings and Operations and Maintenance Manuals	Within 14 days of completion of work	I	CE	CDMS			
31 02 10-1	Water for Dust Abatement	Meter Calibration	At least 21 days prior to use of equipment	A	CE	CDMS			
31 02 30-1	Dust Palliative	Dust Palliative	At least 28 days before beginning onsite work	A	CE	CDMS			
31 03 33-1	Removal and Control of Water	Removal and Control of Water Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 02-3	Compacting Earth Materials	Test Results	Within 7 days after completion of tests	I	CE	CDMS			
31 23 10-1	Earthwork for Site and Structures	Processing Plan for Soils under Structures	At least 42 days before beginning onsite work	A	CE	CDMS			
31 23 10-2	Earthwork for Site and Structures	Test Results	Within 7 days after completion of tests	I	CE	CDMS			

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						CO	CE	TSC	NTUA
31 23 22-1	Pipe Trench Earthwork	Processing Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 22-2	Pipe Trench Earthwork	Test Results	Within 1 day of testing, immediately if test failed	I	CE	CDMS			
31 23 70-1	Controlled Low Strength Material (CLSM)	Approval Data for CLSM Mix	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 70-2	Controlled Low Strength Material (CLSM)	Cold Weather Protection Plan	At least 28 days before beginning onsite work	A	CE	CDMS			
31 23 70-3	Controlled Low Strength Material (CLSM)	Quality Control Test Results	Within 1 day of testing, immediately if test failed	I	CE	CDMS			
31 31 30-1	Soil-Applied Herbicide	Use Plan	At least 28 days before beginning onsite work	I	CE	CDMS			
31 31 30-2	Soil-Applied Herbicide	Applicator Certification	At least 28 days before beginning onsite work	I	CE	CDMS			
31 37 00-1	Riprap	Approval Data and Test Results	At least 45 days before installing rip rap	A	CE	CDMS			
32 15 10-1	Gravel Surfacing	Gravel Certification	At least 14 days before placing surfacing	A	CE	CDMS			
32 15 10-2	Gravel Surfacing	Geotextile Installation	At least 14 days before placing surfacing	A	CE	CDMS			
32 15 10-3	Gravel Surfacing	Geotextile Certification	At least 14 days before placing surfacing	A	CE	CDMS			
32 31 19-1	Steel Ornamental Fence	Certifications	At least 14 days before installing fence	A	CE	CDMS			
32 31 19-2	Steel Ornamental Fence	Installation Plans	At least 14 days before installing fence	A	CE	CDMS			
32 92 20-1	Seeding	Seeding Plan	At least 14 days before seeding	A	CE	CDMS			

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						CO	CE	TSC	NTUA
32 92 20-2	Seeding	Certifications	At least 14 days before seeding	A	CE	CDMS			
33 11 12-1	Steel Line Pipe	Shop Drawings	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-2	Steel Line Pipe	Ventilation Plan	At least 28 days before beginning onsite work	I	86-68141	CDMS			
33 11 12-3	Steel Line Pipe	Field Welder Qualifications	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-4	Steel Line Pipe	Filling and Testing Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-5	Steel Line Pipe	Flotation Prevention Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-6	Steel Line Pipe	Pipe Deflection Measurements	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 11 12-7	***Steel Line Pipe***	***San Juan Lateral Outage***	***At least 28 days before outage***	***A***	***CE***	***CDMS***			
33 13 00-1	Disinfection of Water Utility Distribution	Flushing, Testing, Disinfection, and Verification Plan	At least 28 days before beginning onsite work	A	86-68141	CDMS			
33 13 00-2	Disinfection of Water Utility Distribution	Flushing, Testing, Disinfection, and Verification Report	Within 14 days of test completion	A	86-68141	CDMS			
35 21 95-1	Steel Manifold Piping	Shop Drawings	At least 42 days before construction	A	86-68420	CDMS			
35 21 95-2	Steel Manifold Piping	Commercial Products Data	At least 42 days before construction	A	86-68420	CDMS			
35 21 95-3	Steel Manifold Piping	Final Data	Within one week after testing	A	86-68420	CDMS			
35 21 95-4	Steel Manifold Piping	Final Drawings	Within 28 days of installation	I	86-68420	CDMS			
35 22 14-1	Valves and Equipment	Commercial Products Data	At least 42 days before fabrication or construction	A	86-68420	CDMS			

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						CO	CE	TSC	NTUA
35 22 14-2	Valves and Equipment	Field Installation Procedures	At least 42 days before fabrication or procurement.	A	86-68420	CDMS			
35 22 14-3	Valves and Equipment	Shop Testing Certifications, Reports, and Data	Within 28 days after testing.	A	86-68420	CDMS			
35 22 14-4	Valves and Equipment	Service Manuals and Final Data	At least 28 days before installation	I	86-68420	CDMS			

END OF SECTION

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SECTION 01 51 00
TEMPORARY UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

1.02 REFERENCE STANDARDS

A. Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C2-2023 National Electrical Safety Code (NESC)

1.03 TEMPORARY ELECTRICITY

A. Provide electric power required for construction and testing through contract completion and acceptance by Government.

B. Contractor is responsible for making all arrangements and payments to utility companies for temporary electricity required for construction and testing through contract completion and acceptance by Government.

C. Comply with IEEE C2 clearances and spacing for temporary communications and supply lines.

1.04 TEMPORARY WATER

A. Provide water required for construction through contract completion and acceptance by Government.

B. Potential Sources:

1. Obtain required permits.
2. Depending on availability, the following entities may have potable water:
3. Contact for access, metering, and billing arrangements:
   a. Lower Valley Water Users Cooperative Association
      4286 US-64
      Kirkland, NM 87417
      www.lowervalleywater.com
      Telephone: 505-598-5585
   b. Navajo Agricultural Products Industry

P.O. Drawer 1318
Farmington, NM 87499
www.navajopride.com
Telephone: 505-566-2636
Fax: 505-599-0572

c. Navajo Nation Department of Water Resources / Navajo Nation Water Code Administration
P.O. Box 678
Fort Defiance, AZ 86504
www.nndwr.navajo-nsn.gov, www.watercode.navajo-nsn.gov
Telephone: 928-729-4132
Fax: 928-729-4421

Navajo Tribal Utility Authority (NTUA)
David Scoultz
P.O. Box 170
Fort Defiance, AZ 86504
Telephone: 928-729-6277

d. Reclamation – Four Corners Construction Office (FCCO)
2200 Bloomfield Highway
Farmington, NM 87401
Telephone: 505-325-1794
1235 La Plata Highway
Farmington, NM 87401
Telephone: 505-325-1794

e. CDEC

Gallup, New Mexico Office
Telephone: 505-863-3641

Grants, New Mexico Office
Telephone: 505-285-6656

4. Use water which meets specified requirements for water used in grouting and other permanent work.

C. Arrange, pay, and transport water for use during construction.

1. Provide proof of payment to water utility.

D. Provide means of conveying water to points of use.

1.05 TEMPORARY SANITARY FACILITIES

A. Provide temporary sanitary facilities.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 REMOVAL

A. Remove temporary equipment and facilities upon completion of work under this contract.

END OF SECTION

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SECTION 01 56 32
TEMPORARY SAFETY FENCE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost: Include in prices offered in Price Schedules for other items of work.

PART 2 PRODUCTS

2.01 SAFETY FENCE

A. Fence:
   1. High-density polyethylene grid.
   2. Minimum height: 48 inches.
   3. Color: Safety orange.
   4. Recovered Material Content:
      a. 90 to 100 percent.
   5. Postconsumer Content:
      a. 60 to 100 percent.

B. Posts: Steel fence posts.

2.02 WIRE SAFETY FENCE

A. Metal t-posts with metal grid.

PART 3 EXECUTION

3.01 INSTALLATION

A. Erect fence around work areas at location approved by COR.
   1. Install safety fence as directed by OGR.
   2. Fence along Contractor use areas.
   3. Install temporary safety fencing prior to ground disturbing activities, including surveying. Fencing shall be inspected and accepted by OGR prior to ground disturbing activities, including surveying.
   4. Cultural monitoring and avoidance area fencing shall be installed and maintained.

B. Space posts 10 feet, maximum, on center.

C. Secure grid to posts.

3.02 MAINTENANCE AND REMOVAL

A. Maintain fence until work in area is complete and accepted by OGR.

B. Remove fence when no longer required.

END OF SECTION

SECTION 01 80 15
COMMISSIONING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Commissioning:

1. Payment: Lump-sum price offered in Price Schedules.

1.02 DEFINITIONS

A. Commissioning:

1. Flow testing, component testing, control system testing, power systems testing, verifying component feedback, and verifying operator and control interface with components.
2. Systematic process of ensuring systems and components perform correctly according to design intent and Government's operational needs.
3. Encompasses and coordinates separate functions of system documentation, system and equipment startup, control system calibration, testing and balancing, and performance testing.

B. Commissioning Agent (CA):

1. An independent authority, hired by the Contractor, not otherwise associated with Contractor.
2. Directs and coordinates day-to-day commissioning activities.
3. Does not take an oversight role.

C. Facility: Individual turnout facility (e.g. Coyote Canyon).

D. Lateral Integration Testing:

1. The setup, programming, and testing of the communication link between the facility control system and the San Juan Lateral SCADA System.
2. Major communication link equipment is the Network Interface Panel provided by others. The Network Interface Panel is the communication interface between the San Juan Lateral SCADA System and the facility control system.
3. Performed after installation of the Network Interface Panel and commissioning of the facility.

E. OIT: Operator Interface Terminal.

F. San Juan Lateral SCADA System: SCADA system located at San Juan Lateral Water Treatment Plant that remotely monitors and sends remote control signals to San Juan Lateral facilities

1.03 REFERENCE STANDARDS

A. International Electrical Testing Association (NETA)

1.	ANSI/NETA ATS-2021	Acceptance Testing Specifications for Electrical Power Equipment and Systems

B. International Society of Automation

1.	ISA 62381-2011	Automation Systems in the Process Industry – Factory Acceptance Test (FAT), Site Acceptance Test (SAT), and Site Integration Test (SIT)
2.	ISA 62382-2012	Automation Systems in the Process Industry – Electrical and Instrumentation Loop Check

C. National Fire Protection Association, Inc. (NFPA)

1.	NFPA 70-2026	National Electrical Code (NEC)
2.	NFPA 70E-2027	Electrical Safety in the Workplace

1.04 SUBMITTALS

A. Submit in accordance with Section 01 33 00 – Submittals and Section 01 33 26 – Electrical Drawings and Data.

B. RSN 01 80 15-1, Qualifications

1. Resume.

C. RSN 01 80 15-2, Testing and Commissioning Plan.

1. Detailed list of testing and commissioning activities.
2. Schedule of planned activities.
3. Testing procedures.
    a. Order of required testing and commissioning phases.
    b. Details for each phase, including schedule, flow rates, and procedures to be performed.
    c. Water source used for each phase
    d. Location and method of water discharge for each phase.

D. RSN 01 80 15-3, Testing and Commissioning Reports.

1. Detailed test reports.
2. Issues and resolutions log.
3. Discrepancy and correction list.

E. RSN 01 80 15-4, Post-Testing and Commissioning Documentation.

1. Documentation outlined in section 1.08 Commissioning Documentation Process.

1.05 SCOPE

A. Components supplied, installed, or modified for this Project.

B. At each facility, demonstrate full facility operation through the following, minimum, operating configurations:

1. Full facility start-up and full facility shutdown.
2. Minimum and Maximum facility output flow. Flow data will be provided by the government at the time of commissioning.
3. Demonstrate operation of facility systems and components from the control system graphic displays during these testing activities.

C. At each facility, demonstrate correct operation of components and systems including, but not limited to the following:

1. Control System.
2. Information technology and cyber security.
3. Communication links.
4. Instrumentation.
5. Piping and valve systems.
6. Electrical distribution system.
7. Control and automation functions and operator interfaces used to operate and monitor facility systems with flowing water and without flowing water.
8. At each facility, demonstrate hardware and software functionality, performance, and stability.

D. Perform testing and commissioning activities either required or recommended in accordance with ISA 62381, and ISA 62382 and manufacturer's documentation, including:

1. Field verification of component rating curves and operational setpoints including tuning or adjustment of control loops or settings.
2. Independent verification of flowmeter data.

E. Implement software, settings, and graphic display modifications requested by Government as a result of testing or commissioning activities.

F. Perform and support flow testing under rated flow conditions. Flow data for the flow control valve may be verified with the flowmeter installed upstream of the flow control valve.

1.06 QUALIFICATIONS

A. Commissioning Manager Qualifications:

1. Minimum 10 years of design experience plus 5 years field experience performing installation of mechanical and electrical equipment.
2. Acted as Commissioning Manager on 3 or more facilities of similar size and complexity.

B. Commissioning Manager Responsibilities:

1. Develops the overall commissioning plan.
2. Reviews administrative, construction, pre-commissioning and commissioning programs, and schedules.
3. Recommends and maintains key staff for commissioning process.
4. Coordinates with Government Operations team.
5. Ensures contractor operations involving the control of hazardous energy (lockout/tagout) at Reclamation-owned, -controlled, or occupied facilities and construction sites comply with FIST 1-1, Hazardous Energy Control Program, requirements.
6. Witnesses testing activities, as necessary, in support of the commissioning program.
7. Provides coordination with offsite operating, dispatching, or interfacing agencies, as required.
8. Conditionally accepts equipment and systems for operation during the pre-commissioning testing phase.
9. Collects, reviews, and develops the deficiency lists.
10. Accepts equipment, systems, and facilities, subsequent to successful testing of these items, and provides final acceptance of the project.
11. Operates permanent facility equipment to support the start-up schedule.
12. Refers areas of dispute to Government for final decisions.
13. Monitors control of documentation.
14. Verifies contractual obligations are met.

1.07 COMMISSIONING PROCESS

A. Phases of Testing and Commissioning Work:

1. Including but not limited to the requirements of Contractor Quality Testing and Field Checkout Tests in Divisions 13, 25, 26, 27, 28, 33, and 35.
2. Construction Testing:
    a. Component level acceptance verification testing of components supplied, installed, or modified by Contractor.
    b. Perform in accordance with requirements in these Specifications.
    c. Includes inspections and tests at component and device level.
3. Preoperational Test Phase:
    a. Inspections of complete installation.
    b. Perform site acceptance testing and site integration testing in accordance with ISA 62381.
    c. Perform loop checks per ISA 62382, perform verification for each loop and each input or output.
4. Commissioning (also Operational Test Phase):
    a. Includes cold/dry commissioning and hot/wet commissioning activities.
    b. Technical personnel shall be available onsite to provide consultation with Government personnel if questions or problems arise during testing.
5. Lateral Integration Testing: Setup and testing of communication link between facility control system and San Juan Lateral SCADA System.
6. Reliability Test Run:
    a. Provide minimum 1-day reliability test run at each facility. At completion of commissioning and startup activities:
        1) Coordinate with COR for scheduling reliability test run. Timeframe should be chosen to allow water to be continuously run through the installation.
        2) If a malfunction occurs during reliability test run, restart test run at Contractor's expense.
            a) Malfunction: Failure of system or component to perform satisfactorily which prevents water conveyance for 1 hour or more.
        3) During test run, Government personnel will perform normal system control actions using Contractor supplied equipment.

1.08 COMMISSIONING PROCESS DOCUMENTATION

A. General:

1. Prepare commissioning process documentation to represent a testing program in accordance with guidelines in ISA 62381 and ISA 62382.
2. Inspection and test documentation shall include:
    a. Name and signature of person performing test.
    b. Date at which test was performed.
    c. Identification of component, equipment, or system tested.
    d. Identification of tests performed.
    e. Summary of results of test or inspection activity, including positive identification for each activity of whether system or component passed or failed.
    f. Any data or readings taken during test to verify component, equipment, or system acceptance.

B. Generate the following documentation:

1. Test Plan:
    a. Specific detailed plan for site acceptance and commissioning testing. At a minimum, include:
        1) Detailed list of planned testing and commissioning activities with description of approach used for operating facility during testing and commissioning activities.
        2) Identify systems or components where operation or feedback signals must be simulated during testing or commissioning activities.
        3) Detail anticipated schedule, duration, and sequencing for planned testing and commissioning activities.
2. Test Procedures.
    a. Detailed procedures for planned testing and commissioning activities.
        1) In accordance with specifications and manufacturer's recommendations.
        2) If references are used in testing procedures, include a copy of references used in testing procedures.
    b. If pressurization of system is required in testing procedures or specifications, describe how system will be pressurized, how pressurized water or air will be discharged, and where water or air source will be obtained.
3. Test Reports:
    a. Typed, 8 1/2- by 11-inch sheets.

b. Certified, signed, dated, and attested to having passed requirements and tests.

c. Provide data summary and necessary graphs and plots of tests performed.

d. Available for COR review upon request.

e. Use in preparing final drawings.

4. As-Built Drawings: In accordance with Sections 01 78 30 – Project Record Documents and 01 33 26 – Electrical Drawings and Data.

5. Discrepancy and Correction List:

a. Maintain a list of discrepancies found during inspection, testing, and commissioning activities and how resolved.

6. Testing, Maintenance, and Commissioning Documentation:

a. Master Testing Binder:

1) Contractor assembled and maintained at jobsite.

2) Contains records of testing and commissioning activities performed, including specific test data and reports.

3) Updated as each test or commissioning activity is performed.

4) Available for COR review immediately upon request.

5) Maintain onsite in secure location. Do not remove from site without specific written approval from COR.

6) After completion of testing and commissioning activities, provide to Government. Include copies of test reports and documentation of commissioning activities performed.

b. Maintenance Log:

1) Contractor assembled and maintained at jobsite.

2) Organized by individual component and system.

3) Document Contractor-performed maintenance activities, including associated testing and inspection reports.

4) Maintain onsite in secure location. Do not remove from site without specific written approval from COR.

1.09 COORDINATION

A. Commissioning process requires active participation of Contractor's team, including installation personnel, equipment manufacturers' representatives, and other specific subcontractors (e.g., mechanical, electrical, prefabricated engineered structure), as deemed appropriate.

B. Complete start-up and checkout procedures and ensure readiness of equipment and systems prior to start of functional performance testing phase:

1. Request written confirmation of system readiness for performance testing from appropriate subcontractor, as applicable.
2. Confirm date and times for functional performance testing after readiness confirmation is received for related systems.
3. Provide sufficient time and qualified representatives to complete this process.

C. Coordinate commissioning activities with Government:

1. In accordance with 01 31 19 – Project Management and Coordination.
2. Inform COR in writing of scheduled activities a minimum of 14 days prior to performing activity to allow Government opportunity to witness.
3. Notify COR in writing a minimum of 60 days prior to any activities intended to move water in the installation.
4. Complete each phase of testing and commissioning work and request written confirmation of completion prior to continuing to next phase.
5. Do not schedule testing until procedures are approved.

D. Government will:

1. Witness final functional performance commissioning process.
2. Attest to witnessing final functional performance commissioning process.
3. Final acceptance will be approval of submittal.

E. Government may:

1. Witness inspection, testing, or commissioning activities.
2. Perform additional equipment testing on site with cooperation of Contractor to verify operability of equipment.

1.10 SEQUENCING

A. Complete commissioning requirements of other Sections prior to full facility Operational Test Phase of Commissioning Process.

B. Refer to Section 01 12 10 – Other Contracts and Section 01 14 12 – Work Restrictions and Limitations for additional coordination and scheduling considerations.

C. Do not schedule commissioning activities until test and commissioning procedures are approved.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 GENERAL

A. Provide instrumentation, test equipment, and personnel necessary for performing tests.

3.02 EQUIPMENT SPECIFIC TESTING REQUIREMENTS

A. General:

1. These requirements are in addition to those listed elsewhere in Specifications and Contractor-developed testing and commissioning documentation.
2. Meet equipment manufacturer performance requirements.
3. Inspect equipment and components for dirt, foreign material, and damage to component or coatings.
4. Flush and replace lubricants where required by manufacturer or by condition of lubricating material.
5. Check and correct leveling plates, bearing plates, anchorage devices, fasteners, and alignment of components.
6. Operate valves and other devices.
7. Equipment shall function as specified before the Government will consider the Facility Commissioning phase complete. If equipment does not operate as specified, make required adjustments and restart the test.

B. Control System:

1. Demonstrate accurate operation of graphic displays and operator interfaces to graphic displays and OIT. Operate field components and instruments for operating configurations and signals represented on OIT and graphic displays to very accuracy of feedback signals.
2. Test each setpoint and control point.
3. Operate field components and instruments for operating configurations and signals represented on OIT and graphic displays to verify accuracy of feedback signals.
4. Verify system alarms:
    a. Ensure any alarms operate as intended.
    b. Where simulation of feedback signal is required to verify an alarm, perform simulation to verify system interlock unless device is inaccessible.
    c. Provide feedback by operating or forcing from device, if it does not damage device or present safety hazard.
5. Check alarms and annunciation points.

6. Adjust and tune control loops, logic, settings and OIT, and graphic displays.
7. Flow Control Valve (FCV) solenoid pilot valve operation:
    a. Ensure solenoid pilot valve repositions upon application and removal of control power from Control System.
    b. Confirm that the FCV limit switch status indications transmitted to the Control System OIT are accurate. The OIT graphical display shall correctly reflect valve position from limit switch feedback.
8. Verify logic and operation of system.
9. Demonstrate operation under each control mode.

C. Information Technology and Cyber Security:

1. Capture NetFlow data between components for network and protocol analysis at time of system commissioning.
2. Perform vulnerability scan and correct deficiencies.
3. Verify account-based access control.
4. Validate network architecture including port address lists and device address lists.
5. Validate and document communication restrictions, filtering, and monitoring rules through firewalls and other network components.
6. Perform additional Contractor- and manufacturer-recommended testing.
7. Government may perform additional cyber security related testing and verification of system with cooperation from Contractor.

D. Communication Links:

1. Verify cable is free of defects and meets design and manufacturer's guidelines.
2. Verify communication links between devices.
3. Demonstrate feedback signals are transmitting correctly and are operating correctly between facilities.

E. Instrumentation:

1. Inspect each gauge, sensor, transmitter, thermometer, and other instruments. Replace defaced or broken items.
2. Inspect instrument tubing.
3. Field verify calibration and performance of each instrument. Document results for each instrument on separate field calibration sheet. Include, at minimum:
    a. Input, output, and error at 0, 25, 50, 75, and 100 percent of span for analog instruments.
    b. Switch setting, contact action, and deadband, if applicable, for discrete elements.

4. Flowmeter Simulated Tests
    a. Apply simulated flow signal inputs. Simulated signals to cover minimum and maximum flows over the velocity range.
    b. Check displays for correct readouts and check each I/O point for correct I/O signal.
    c. Test each output connection for correct output conformance.
    d. Verify instrumentation and indication devices function properly. Test functions of flowmeter transmitter for correct operation.
5. Required I/O and feedback data not visible on OIT shall be demonstrated by temporarily connecting a computer to the PLC to verify the data is available to be transmitted to the SJLWTP SCADA System.
6. Correct failures detected during loop checks prior to cold/dry and hot/wet commissioning activities.

F. Valves:
1. Inspect valves, and clean bonnets and stems.
2. Check valve orientation and operation of position indicators.
3. Verify valve seats are free of foreign matter and positioned for intended service.
4. Adjust until valves are bubble tight, and valves do not leak water past seats.

G. Piping Systems:
1. Verify piping system is clean and free of foreign matter before filling.
2. Tighten flanges and other pipe joints after system has been placed in operation.
3. Replace gaskets that have been previously compressed or show signs of leaking.
4. Inspect joints for leakage:
    a. Promptly remake each joint that appears to be faulty; do not wait for corrosion to form.
    b. Clean threads, apply compound, and remake joints.
5. After system has been placed in operation, clean strainers, drives, pockets, orifices, valve seats, and headers in fluid system to ensure freedom from foreign matter.
6. Inspect insulation and replace where damaged.

H. Electrical Distribution System:
1. Perform equipment testing in accordance with NETA ATS and Division 26 specifications.
2. Testing shall be limited to: visual alignment, mechanical operation, bolt-torque verification, contact resistance, and insulation-resistance (megger) testing of the

main bus bars. Check and document settings for electrical protection and metering components.

3. Testing for simple electrical power systems shall include the following at a minimum:
    a. Verify correct installation of the main panel, main breaker, and branch breakers.
    b. Verify torque of terminals per manufacturer instructions.
    c. Perform continuity and polarity checks on branch circuits.
    d. Verify correct operation of lighting circuits and switches.
    e. Verify dedicated circuits for fire alarm control panel, fire suppression equipment, and cell booster power are installed per plans and remain energized from the main panel.
    f. Perform receptacle testing including GFCI function where required.
    g. Verify labeling of breakers and circuits.
    h. Record panelboard schedules “as-left” and confirm alignment with field installation.

I. Flow Testing:
1. Operate flow control valve to verify proper operation of components, condition of components, and documentation of any leakage or vibrations.
2. Demonstrate ability of facility to operate within designed operating range. Include operation at maximum and minimum facility output flows.
3. Full facility start-up and full facility shutdown.
4. Demonstrate operation of facility as detailed in Article 1.05 B5.
5. Operate facility systems and components from the Control System graphic displays during the flow testing activities.
6. Refer to Annex A of ISA 62382 for guidance on testing of hydraulic equipment during commissioning.
7. Simulate abnormal operations and failures of instrumentation and actuators/valves:
    a. Simulations include but are not limited to:
        1) Loss of power.
        2) Loss of instrumentation signal.
8. Government:
    a. Will provide general guidelines for operating points to test and time interval to test.

b. May perform additional testing and verification of system with cooperation from Contractor.

9. Contractor:

a. Specific test procedures.

b. Onsite technical personnel for support during testing.

c. Provide Master Test Procedure Binder prior to start of commissioning. Include:

1) Comprehensive set of test procedures and test plans for testing, commissioning, and inspection activities required in this specification or in other specifications in this contract package.

2) Schedule and sequencing of testing activities.

d. Test reports:

1) Typed.

2) Certified, signed, dated, and attested to having passed requirements and tests.

3) Include data summary and necessary graphs and plots of tests performed.

4) Available for COR review upon request.

J. Lateral Integration Testing

1. Test communication links between the Turnout Building Control System and the San Juan Lateral SCADA System.

2. Minimum duration of lateral integration testing per facility: 3 days.

3. Government will witness test. Notify COR in writing at least 20-days prior to beginning of Lateral Integration Testing. Provide individual notification to COR for each Turnout Building.

4. Provide the following minimum technical personnel to support integration with the lateral communication system. Technical personnel shall be onsite throughout the entire duration of the lateral integration testing.

a. Two electricians.

b. SCADA/Control System representative capable of performing onsite modifications to the Control System.

5. Schedule for Lateral Integration Testing:

a. Perform Lateral Integration Testing after installation of the Network Interface Panel, which is installed by Others in accordance with Section 25 00 01 – System Control and Monitoring. Allow a minimum range of four weeks of onsite access to allow installation of the Network Interface Panel.

b. Perform Lateral Integration Testing after other commissioning activities for the Turnout Building site are complete, this includes flow testing and online testing activities.

c. Coordinate Lateral Integration Testing with the COR and the San Juan Lateral Water Treatment Plant Contractor. Section 01 14 12 – Work Restrictions and Limitations contains timeframe and schedule limitations for the conduct of Lateral Integration Testing.

END OF SECTION

SECTION 25 00 01
SYSTEM CONTROL AND MONITORING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. System Control and Monitoring:

1. Payment: Lump sum prices offered in the Price Schedules.

1.02 DEFINITIONS AND ACRONYMS

A.	FIT:	Flow Indicating Transmitter.
B.	GFE:	Government Furnished Equipment.
C.	HART:	Highway Addressable Remote Transducer.
D.	I/O:	Input/Output.
E.	LCD:	Liquid Crystal Display.
F.	LED:	Light-Emitting Diode.
G.	NTUA:	Navajo Tribal Utility Authority.
H.	NGWSP:	Navajo-Gallup Water Supply Project.
I.	OIT:	Operator Interface Terminal.
J.	P&ID:	Piping and Instrumentation Diagram.
K.	PLC:	Programmable Logic Controller.
L.	PoE:	Power-over-Ethernet.
M.	RTD:	Resistance Temperature Detector.
N.	SCADA:	Supervisory Control and Data Acquisition.
O.	SJLWTP:	San Juan Lateral Water Treatment Plant.
P.	UPS:	Uninterruptible Power Supply.
Q.	VPN:	Virtual Private Network.

1.02 REFERENCE STANDARDS

A. Electrotechnical Commission (IEC)

1.	IEC 61131-3-2025	Programmable Controllers – Part 3: Programming Languages

B. Institute of Electrical and Electronics Engineers (IEEE)

1.	IEEE 802.3-2022	Local and Metropolitan Area Networks - Specific Requirements Part 3:Carrier Sense Multiple Access with Collision Detection Access Method and Physical Layer Specifications
2.	IEEE 1815-2015	Electric Power Systems Communications – Distributed Network Protocol (DNP3)

C. International Society of Automation (ISA)

1.	ISA 101.01-2015	Human Machine Interfaces for Process Automation Systems
2.	ISA 62443-1-1(99.01.01)-2007	Security for Industrial Automation and Control Systems Part 1-1: Terminology, Concepts, and Models
3.	ISA 62443-2-1(99.02.01)-2009	Security for Industrial Automation and Control Systems Part 2-1: Establishing an Industrial Automation and Control Systems Security Program

D. National Electrical Manufacturers Association (NEMA)

1.	NEMA 250 - 2014	Enclosures for Electrical Equipment (1,000-Volts Maximum)
2.	NEMA BI 50009-2024	Low Voltage Cartridge Fuses
3.	NEMA IA 10030-2024	Industrial Control and Systems: Enclosures
4.	NEMA IA 10039-2025	Industrial Control and Systems: Control

E. National Fire Protection Association, Inc. (NFPA)

1.	NFPA 70-2023	National Electrical Code (NEC)

F. National Institute of Standards and Technology (NIST)

1.	NIST SP 800-82 Rev2	Industrial Control Systems (ICS) Security

G. Underwriters Laboratories, Inc. (UL)

1.	UL 508-1999	Industrial Control Equipment, Seventeenth Edition

1.03 SUBMITTALS

A. Submit the following in accordance with Sections 01 33 00 – Submittals and 01 33 26 – Electrical Drawings and Data.

B. Drawings and data in sufficient detail to show full compliance with this specification and applicable codes.

C. RSN 25 00 01-1, Qualifications – Contractor, Designer, Programmer, and Installer work summary, include:

1. Description of projects and experience intended to be used to meet experience requirements listed in Article 1.04 Qualifications.
2. Include name, date, and location of projects.

D. RSN 25 00 01-2, System Description Documentation:

1. Functional System Description:
    a. Describe and document the proposed Control System, its features and limitations, compliance with the requirements, operation, software algorithms, display screens, and software configuration.
2. System overview:
    a. Overview of hardware, software, control functions, interfaces, and general data flow associated with entire system.
    b. Include software block diagram of entire system.

E. RSN 25 00 01-3, Approval Drawings and Data:

1. Drawings:
    a. Instrumentation Loop Diagrams, Schematic Diagrams, and Wiring Diagrams:
        1) Show all wiring and cable connections for supplied components, including all connections made inside cabinets, panels, or other enclosures.
        2) Include external connections showing all connections to Control System components, including components required under other Specifications. Identify the remote end of the cable connections on the drawings including identifying the name and tag of the connected device as well as the terminals used for the connection.
    b. Networking Diagram.
        1) Show all communications links, including, but not limited to servers, network switches, media converters, and communication links between the PLC, OIT, and other field devices.

c. Provide in accordance with the requirements for Computer drafted drawings in Section 01 33 00 – Submittals.

2. Bill of Materials.

3. Manufacturer's product data and catalog sheets on proposed system components.

a. Identification of model or catalog number proposed, equipment capabilities, and electrical characteristics.

4. Uninterruptible power supply calculations to meet requirements found in Article 2.13 E.

5. Spare part list.

F. RSN 25 00 01-4, Layout Drawings:

1. Dimensioned drawings showing relative positions of components. Includes component identification and layout information.

2. Enclosure and Control Panel Layout Drawings. Exterior and interior arrangements.

3. Include nameplate information.

4. Provide in accordance with the requirements for Computer drafted drawings in Section 01 33 00 – Submittals.

G. RSN 25 00 01-5, Software Development Documentation:

1. Include the following for the Programmable Controller:

a. I/O List, including at a minimum the following information:

1) Point designation.

2) Point description.

3) Card location.

4) Type and rating information.

5) Termination data.

b. Data points list for soft data points internal to programmable controller.

1) Includes information on logical and calculated datapoints, as well as datapoints that are communicated between components.

2) Point designation.

3) Point description.

4) Type and rating information.

5) Address.

c. Communication link information.

d. Graphic display drawings (proposed LCD display and OIT touch panel) for each screen available on the LCD or OIT displays.

e. Logic diagrams: Functional and binary logic drawings for the entire Control System.

f. Manufacturers data on all software used and any software licenses provided for the project. Includes all software applications used on or for the OIT, PLC, etc.

H. RSN 25 00 01-6, Final Drawings, Documentation, and Operations and Maintenance Manuals:

1. All information supplied should be final documentation and should include all "as-built" changes including any changes made through the Lateral Integration Testing referenced in Section 01 80 15 – Commissioning.

2. Final drawings:

a. Includes final versions of all previously submitted diagrams and drawings with as-built changes incorporated into them.

b. Layouts, schematic diagrams, and wiring diagrams including "as-built" changes.

3. Final version of approved software documentation including "as-built" changes.

4. Operations and Maintenance Manuals:

a. Include operation of following system features and hardware items:

1) Screen displays and control operations.

2) General description of all control actions.

3) Limit and trend settings.

4) Manual status entry.

5) Report generation.

6) Control access methods.

5. Final I/O point list.

6. Final uninterruptible power supply calculations to meet requirement found in Section 2.13 E.

7. Final spare parts list.

8. Handwritten updates or any updates that do not meet the requirements for computer drafted drawings in Section 01 33 00 – Submittals are not acceptable. Drawing modifications must be made in the program originally used to generate the drawing and modifications made by a pdf editing program, like Adobe Acrobat or Bluebeam, are not acceptable.

1.04 QUALIFICATIONS

A. Contractor:

1. Minimum 7-years' experience with similar projects including systems with programmable controllers.
2. Completed at least 5 projects of similar size, type, application, and complexity. At least three projects must have included using proposed Schnieder Electric programmable controller product line, software, and graphics package

B. Designers and Programmers:

1. Minimum 5-years' experience programming systems of similar size and complexity containing programmable controllers.
2. Completed at least 2 projects of similar size and complexity using proposed Schnieder Electric programmable controller product line, software, and graphics package

C. Installer:

1. Minimum 5-years' experience installing control panels and control equipment.
2. Minimum 5-years' experience performing any testing or commissioning activity installer is expected to perform.

1.05 DESIGN REQUIREMENTS

A. General:

1. Provide a Control System for each of the following facilities:
    a. Sanostee Turnout.
    b. Newcomb Turnout.
    c. Naschitti Turnout.
    d. BBN9 Turnout.
    e. Coyote Canyon Turnout.
    f. Rock Springs Turnout.
    g. Window Rock Turnout.
2. Control System shall be the primary control, automation, and monitoring system for instruments and equipment located in each Turnout Building and on the Turnout Building site:
    a. Provide the Control System with associated equipment, software, terminations, and wiring to implement control and monitoring functions described in these Specifications.
    b. Design Control System with functional bi-directional data links to other systems, components, PLCs, PACs, and networks as required.

c. Control System shall provide support for the following minimum communication protocols:

1) DNP 3.0 in accordance with IEEE 1815.

2) Modbus Serial (RTU and ASCII).

3) Modbus Network (TCP).

4) HART Communications Protocol.

d. Fully integrated system, including the following minimum features:

1) Processing units, I/O components, network and communication components, power supplies, and other miscellaneous control components.

2) Graphic displays viewable from OIT.

3) At minimum, I/O identified in Table 25 00 01A – Typical Turnout Building I/O and Data Point List, and the P&ID shown in drawings 1695-D-62569 through 1695-D-62577.

e. Provide Control System programming, control logic, and macros:

1) Includes development of logic diagrams, flow charts, and OIT graphical displays.

2) Provided documentation and descriptions listed within software, including descriptions for each rung, block, or line of ladder diagrams, function block diagrams, and structured text logic.

f. Provide interface design:

1) Include development of schematics, datalinks, and other applicable items, required for interfaces between components.

3. Provide Programmable Controller with required I/O and local control at the Turnout Building:

a. Provide media converters, network switches, and fiber optic patch panels as required to complete installation.

b. I/O components, communication components, backplanes, and other components utilized in the Control System shall be of the same manufacturer. Minimize number of different types and models utilized.

c. Failure of remote I/O hardware or network communication link must be individually alarmed at the OIT in the Turnout Building.

4. Software packages used in the installation and for development of system graphics, control logic, shall be up-to-date and compatible with the most recent version of major operating systems, including Microsoft Windows. Provide software not mentioned specifically, but essential for system to meet functional requirements.

5. Control System shall interface to a long-haul fiber optic network at each site, which is provided either by NTUA (utility) or Government (underground).

6. Provide additional equipment not specified in this section but necessary to make system complete and operational at no cost to Government.

B. Control System Function:

1. Utilize the long-haul optical fiber transmission lines to communicate telemetry, monitoring, and control data from the Turnout Building Control System to the SJLWTP SCADA system.

2. Monitoring and Alarming:

a. Monitor the instruments and equipment in the Turnout Building and record and display local telemetry and monitoring information.

b. Display information from devices or systems on graphic displays. Display information in units of process measurement where applicable.

c. Monitor and provide alarming for instruments and devices shown on the supplied P&IDs 1695-D-62569 through 1695-D-62577, the I/O table of Article 3.01 of this Specification, and where the Specifications reference interfaces with PLCs, facility control, or SCADA systems.

d. Status alarms for Control System equipment malfunctions.

e. Where Flow Indicating Transmitter (FIT) devices are shown on P&IDs, display the following minimum information on graphic display:

1) Flow direction.

2) Flow rate.

3) Total flow.

3. Control:

a. Where solenoid operated flow control valve is shown on P&ID, display the status of the flow control valve on graphic display.

1) Flow Control Valve Operating.

2) Flow Control Valve Closed.

C. Remote Interface:

1. Provide remote interface from the Control System at each site to the remote San Juan Lateral SCADA System (the San Juan Lateral SCADA System is by Others). The Control System shall transmit and receive monitoring and control data to and from the San Juan Lateral SCADA System. Provide system programming, logic development and register address assignment as necessary to complete the remote communication link. Interface will be over the long-haul optical fiber transmission network connection.

a. Contractor shall coordinate with BOR for setup of interface signals between the facilities and San Juan Lateral SCADA system.

2. Long Haul Optical Fiber Transmission Lines:

a. For the Sanostee, Newcomb, Naschitti, BBN9, and Coyote Canyon Turnout Buildings, an NTUA (utility) provided long-haul optical fiber transmission system shall be the long-haul transmission lines that are used for the long-haul optical transmission network connection to the SJLWTP SCADA System at each site.

b. For the Rock Springs and Window Rock Turnout Buildings, a Government provided underground long-haul transmission lines that are used for the long-haul optical transmission network connection to the SJLWTP SCADA System at each site.

c. A Network Interface Panel ***will be provided by Others for each site. The Network Interface Panel w***ill be used to interface to the long-haul optical fiber transmission lines, as shown on drawing 1695-D-62588.

1) The Network Interface Panel shall be utilized by the Control System to connect the respective long-haul optical fiber transmission line network via the Network Interface Panel network switch and fiber bypass unit.

2) Network Gateway switch in Network Interface Panel will act as the firewall between the Control System and the long-haul communication network.

3) Contractor to coordinate with Government on installation of Network Interface Panels and power and communication connections.

d. Primary communication system utilized for transmission of the following signals:

1) SCADA/Control Data: Modbus TCP communication protocol.

3. Remote Interface Communication Link:

a. Design and provide communication link between the Network Interface Panel, which will be provided by Others, and Contractor supplied Control System equipment.

1) ***Provide power and communication interface for the Network Interface Panel, this includes providing all circuit breakers, fuses, terminal blocks, patch panels and other components necessary for the connection to the Network Interface Panel and should be ready for the installation of the cables to connect to the Network Interface Panel.***

2) Cables connecting the Network Interface Panel to the Contractor supplied equipment will be provided by Others.

b. Contractor is responsible for programming, setup, and testing required for Contractor supplied equipment, including the Contractor supplied PLC to support the communication link. The SJLWTP Contractor will program the Network Interface Panel communication equipment.

c. Control data leaving and entering the Control System site shall be secured utilizing IPSec VPN tunneling. Control data includes data passed through Programmable Controllers. The IPSec VPN shall be implemented by the network interface switch located in the Network Interface Panel.

D. Graphic Displays

1. Show monitoring feedback and status of components and system processes tied into control and monitoring system.

2. View graphic displays from OIT.

E. Provide Door/Gate/Hatch Intrusion Monitoring:

1. Utilize balanced magnetic contacts to provide door, gate, control panel, network interface panel, and security and communication network rack intrusion monitoring.

2. Provides specific indication of which entry point opened.

3. Minimum Locations:

a. Doors or entry points for entering the building. This includes personnel doors, equipment doors, and roll-up doors.

b. Equipment control panels and network racks.

c. Gates allowing entry to the site or through fences.

4. Alerts: Provide hardwired (not network or communication based) alarms to Control System for system alarm or unauthorized entry.

F. Power:

1. Provide power distribution to Control System components, including protection, to support full operation of system. This distribution system shall include analog loop power supplies and contact interrogation voltages for dry contact inputs.

2. ***Deleted***

G. Software:

1. Software, including graphics and control logic implementations shall be configurable by Government to allow for future modifications to Control System:

a. Graphic display configuration software shall include libraries of standard displays, nameplates, and components utilized by Government for future Control System modifications

b. Modules used in control logic implementation shall be viewable and configurable by Government, including internal implementation of those modules.

2. Password protect access to PLC program loading. Provide the following access levels in the software:

a. Status access:

1) View only access to facility displays and trends.

2) No access to process control or alarm acknowledgment.

b. Operator access:

1) Access to facility displays, trends, process control, and alarm acknowledgement.

c. Administrator:

1) Access to facility displays, trends, process control and alarm acknowledgement.

2) Access to view and modify Control software.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Handle and store hardware in accordance with manufacturer's instructions. Include copy of manufacturer's instructions with equipment at time of shipment.

B. ***Deleted***

1.07 EXTRA MATERIALS

A. Furnish a set of special tools and accessories required for installation, normal operation, and maintenance for each facility. Include:

1. Contractor and manufacturer recommended tools and accessories.

2. Tools and accessories recommended for field maintenance and those designed to aid in the diagnosis, maintenance, repair, and replacement of supplied equipment.

B. Furnish spare parts. Furnish a set of the following components:

1. I/O Cards: Minimum of 10 percent spare of each type used, no less than one (1) of each size and type used.

2. Network and Communication Cards or Components: Minimum of one (1) spare of each type of control system component.

3. Programmable Controllers: One (1) fully redundant spare set of automation controllers.

4. Back planes: Minimum of one (1) spare of each type of control system component.

5. Communications Converters: Minimum of one (1) spare of each type of communications converter used.
6. Miscellaneous Components: Minimum of one (1) spare of each type of control system component.
7. OIT: One spare of each type.
8. Contractor or manufacturer recommended spare parts.
9. Replace spare parts used during the installation, testing, and commissioning process to provide a complete set of spare parts.

C. Furnish two copies of system software. Include:

1. Software required to re-install software required for the system to function properly, for each facility.
2. Copies supplied shall contain the final version of furnished software.

PART 2 PRODUCTS

2.01 PROGRAMMABLE LOGIC CONTROLLER

A. Modicon M340 Programmable Controller including accessories shall meet the following requirements:

1. Chassis: DIN rail mountable.
2. 24 VDC Power Supply.
3. Modbus and Ethernet capable.
4. Provide with a minimum of 40 percent spare capacity, including memory utilization, processing capacity, and other functions.

2.02 OPERATOR INTERFACE TERMINAL (OIT)

A. Provide OIT to perform as primary operator interface to the Control System. Harmony ST6 Touch Screen Panel as manufactured by Schnieder Electric, or similar.

1. Mount on Control System Enclosure.
2. Color TFT LCD. 15-inch touch screen display, minimum.
3. Modbus TCP/IP, RS485, RS232.
4. EcoStruxure Operator Terminal Expert Software.

2.03 INPUT/OUTPUT (I/O)

A. Provide I/O as required to meet specified functionality.

1. Provide I/O cards intended for use with the Modicon M340 product line, manufactured by Schneider Electric http://www.schneider-electric.com.

2. Provide 10 percent spare installed I/O of each size and type at each I/O location.

3. Discrete and Analog Input and Output modules as required to implement monitoring and control functions described.

4. Provide capability to install 20 percent additional future I/O cards at each I/O location. Includes connection capability, space on backplanes and sufficient margin in power and operating capacity for a true 'plug and play' installation of additional I/O cards.

2.04 PROGRAMMING CONTROL LOGIC

A. Provide capability of programming control logic in Control System using the following programming languages and in accordance with IEC 61131-3:

1. Ladder Diagram.

2. Function Block Diagram.

3. Structured Text.

B. Implement majority of control logic in Control System using Ladder Diagram or Function Block Diagram, with only select areas implemented using other methods if necessary.

C. Emphasize readability and maintainability in control logic implementation.

D. Fully document control logic developed for Turnout Building.

2.05 GRAPHIC DISPLAYS

A. Provide graphic displays for each Control System in accordance with requirements and recommendations in ISA 101.01. ISA 101.01 recommendations and requirements in addition to requirements included in this Specification represent minimum level of functionality for graphic displays.

B. Control System graphic displays are subject to approval by COR.

C. Spare capacity for an increase of up to 20 percent in the number of graphic pages.

D. Designed to support user situational awareness, intuitive navigation, and intuitive system status and condition information.

E. Designed to support types of tasks and activities required of the operator. Control of system and modification of operational setpoints shall normally be accomplished through OIT.

F. Separate graphic display screens shall be designed and provided for screens with different functions and locations.

G. Use consistent design philosophy across graphic displays, however, the specific functions desired for the individual screens may require unique arrangements of graphics and information.

H. Control System graphic displays shall incorporate a hierarchy of displays that provide progressive exposure to detail. The following list provides a general guideline for the hierarchy of displays:

1. Highest level display shall show entire system within operator's control. It is intended to act as an overall indicator as to how the site is running by tracking key performance indicators.
2. Displays shall consist of detailed information on system operations.
3. Equipment operation if not included in system operation.

I. User interaction techniques:

1. Clear, with graphics that are consistent in the method that they present similar information.
2. Graphic functions: Standardized, intuitive, and straight forward with minimal user manipulation.
3. Error tolerance in user interaction for critical devices shall be included, with simple notification of error and effective methods for recovery.

J. Develop a color scheme for graphic displays for approval by COR:

1. Grey backgrounds for graphics, with minimal use of color in graphics.
2. Utilize color very specifically and consistently.
3. Color alone shall not be the sole differentiator of an important condition or status.

K. Include analog representation of important measurements, indicating their value relative to normal, abnormal, and alarm conditions.

L. Include visual indication of an Operator input field, or other operator-initiated control function, has been disabled.

M. Response time requirements for the graphic displays:

1. Graphic displays shall take no longer than 3-5 seconds to appear when called up.
2. Update live information on display every 2-seconds unless critical nature of process information displayed requires more frequent updating.

N. Provide feedback on the completion of an action or the failure to complete an action.

O. Component Depiction:

1. Utilize standard shapes for components. Shapes shall be simple and avoid unnecessary detail.

2. Show status of components clearly and consistently.
3. Include operating hours on operating and standby rotating equipment.
4. Provide faceplates to display detailed component information and user input fields.
5. Vessels and Tanks: Size shown shall be relative to process importance of component and relate to its physical size.
6. Process Lines: Limited number of line types and line thickness.
7. Process Controllers:
    a. Depicted as a physical entity on the graphics, with key values like set point, output, and control mode displayed.
    b. Provide faceplate type displays for process controllers with detailed information and user input fields, where required, for the process controller.

P. Text and Values:
1. Do not display leading zeros when depicting values unless the value is less than 1.
2. Differentiate live values from static text, for example by depicting live values with a color. Units of measure shall not be depicted as a live value.
3. Date and time display:
    a. Display date in form of "YYYY-MM-DD".
    b. Display time of day in form "hours:minutes:seconds". Update time display automatically once every second.

Q. Page title: Each format shall have appropriate page title.

R. Provide visual display of lockout/tagout application.

S. Provide faceplates for sequences, including automatic startup and shutdown.

T. Annunciate active trip conditions on the graphic.

U. Alarms:
1. Alarms and indications of abnormal situations shall be clear, prominent, and consistently distinguishable.
2. Provide the following minimum alarm displays:
    a. Alarm status.
    b. Alarm log.
3. Displays shall provide the Operator the ability to acknowledge alarms.
4. Place alarms out of service (access-controlled operation).
5. Modify alarm attributes (access-controlled operation).

6. Alarm indication in the Control System shall have audible alarms associated with them. Sound level of the alarm shall be enough for easy detection.

V. No animation except for specific alarm related graphics.

W. Alarm and Events:

1. Maintain a consistent approach to alarming across Turnout Buildings.
2. Allow for alarm monitoring for discrete and analog inputs/outputs and communication links.
3. Document the following for each alarm:
   a. Alarm type.
   b. Alarm setpoint.
4. Indicate acknowledged alarms, unacknowledged alarms, and unacknowledged alarms which have returned to normal.
5. Time stamped logs for alarm history, event history, diagnostic, and audit history for the alarm system.

2.06 CONTROL

A. Designed to aid operator over complete operating range of the facility while controlled equipment is in service. Available and configurable parameters useable to control equipment within manufacturer's limitations and enhance equipment life shall be incorporated into the system design.

B. Provide modifiable setpoints and control variables, including alarm setpoints, delays, and control setpoints that are modifiable by operator at OIT. Sufficient setpoints shall be included to allow operator adjustment of system through desired operating range. Values referred to as 'setpoints' in these Specifications shall be modifiable from OIT.

C. Provide control logic to mitigate bad quality or out of range signals. Logic shall drive devices to a safe state, substitute a value, or hold last value as appropriate for application.

D. Fault checking shall be included and cause appropriate response, change over, or alarm.

E. Upon initial system power up or system power restoration, systems shall be in manual with outputs in "off" or fail-safe mode.

F. Provide rapid shut down and emergency shutdown of equipment.

G. Make appropriate adjustments to Control Systems for areas where actual Contractor supplied equipment and installation varies from the information included in Contract Documents.

H. Flow Control Valve Control:

1. Provide Flow Control Valve control from Control System in accordance with Section 35 22 14 – Valves and Equipment.
    a. When the Control System issues a close command to the Flow Control Valve, the Flow Control Valve solenoid shall be energized to cause the valve to close.
    b. When the Control System removes it command to close the Flow Control Valve, the Flow Control Valve solenoid will be deenergized to allow Flow Control Valve normal operations.

I. Monitoring and Alarming:

1. Where monitoring or alarming are required in these Specifications at a minimum provide the following functionality. This does not preclude additional functionality required by other requirements in these Specifications or required system functionality.
2. Monitoring
    a. Display information received from end devices on graphic displays. Display information in units of process measurement where applicable.
    b. Where Flow Indicator Transmitter (FIT) devices are shown on P&IDs, display the following minimum information on graphic displays:
        1) Flow Direction.
        2) Flow.
        3) Total Flow.
3. Alarming:
    a. Monitor and provide alarming for instruments and devices shown interfacing with the Control System Enclosure on the supplied P&ID drawings, Table 25 00 01A – Typical Turnout Building I/O and Data Point List, and where the Specifications reference interfaces with PLCs or Control System:
    b. Provide status alarms for Control System equipment malfunctions.

J. Turnout Building Operation:

1. Operational logic should incorporate data-points and system inputs shown in Table 25 00 01A – Typical Turnout Building I/O and Data Point List and supplied Process and Instrumentation Diagrams.
2. Additional Facility Monitoring:
    a. Monitor access to monitoring system in accordance with Section 28 10 01 – Security and Surveillance System.
    b. Status alarms for UPS system(s).
    c. Status alarms for Control System equipment malfunctions.

K. Remote System Interfaces:

1. Provide the following remote-control signals for interface with the future San Juan Lateral SCADA System as shown on Table 25 00 01B – SJLWTP SCADA System Remote Interface Data Point List. Provide system programming, logic development and register address assignment to implement remote control interface to complete the remote communication link.
2. Interface will be via Modbus TCP communication over the long-haul optical fiber network connection:
    a. Control System Local/Remote Control.
        1) Local Control shall allow operator in Turnout Building to operate Control System from OIT.
        2) Remote Control shall allow the San Juan Lateral SCADA System to operate the equipment.
    b. Automatic Operation Control Signals:
        1) Flow Control Valve Close.
        2) Flow Control Valve Not Closed.
3. Contractor shall coordinate with Government and San Juan Lateral SCADA System installer for setup of interface signals between the two systems.

L. Safety Measures:

1. Failure of PLC, remote I/O hardware, or network communication link shall be individually alarmed at OITs.
2. Watchdog function to monitor:
    a. Internal processor clock failure.
    b. Processor memory failure.
    c. Loss of communication between processor and I/O modules.
    d. Processor ceases to execute logic program.
3. Initial safety conditions:
    a. Utilize program module to dictate output states in known and safe manner prior to running of control program.
    b. Utilize program each time PLC is re-initiated and control program activated.
4. Monitoring of internal faults and display:
    a. Internal PLC system status and faults shall be monitored and displayed:
        1) Monitored items shall include:
            a) Memory ok/loss of memory.
            b) Processor ok/processor fault.

c) Scan time overrun.

5. Operator intervention:

a. Logic system failure shall not preclude proper operator intervention:

1) Safety shutdown of equipment or system shall require manual operator intervention before equipment or system operation is reestablished.

2.07 LOCKOUT/TAGOUT

A. Lockout/Tagout functionality shall provide the operator with the following minimum control features:

1. Ability to lockout individual components.

2. Ability to lockout process systems.

B. Upon application of a lockout/tagout command system shall automatically stop, or remove from service, affected component or system. This operation shall be performed regardless of control mode that system is in:

C. Control System shall include logic to automatically react to the application of lockout/tagout on a component, or system, by making appropriate adjustments or shutting down impacted systems.

2.08 USER ACCESS CONTROL AND MONITORING SECURITY

A. Provide the following minimum access control capabilities, in accordance with recommendations in ISA 62443-1-1, ISA 62443-2-1 and NIST SP 800-82:

1. Configurable account password solution to include requirements for password length, frequency of change, minimum required password complexity, number of login attempts, inactive session logout, screen lock by application, and denial of repeated or recycled use of the same password. Adjustments to these requirements require explicit written approval of COR upon receipt of written justification.

2. Network Components: Provide individual accounts for each user with customizable access privileges for each account. Accounts shall be configurable into role-based functions allowing varying levels of interface with and modifications to the system components. Include account authentication by approved method.

B. Prior to testing of Control System and components, remove or disable software components that are not required to provide specified system functions and capabilities or required for operation or maintenance of the system.

C. Provide logging of all events, including access events for network components. Forward logs to central server for long term storage.

D. Provide operating logs for windows operating systems.

E. Provide approved anti-virus protection on components where applicable.

2.09 NETWORK

A. Provide high speed data network between automation controllers and processors:

1. Network shall be designed to be scalable and facilitate future expansion of the system.
2. Network design shall allow any component associated with a given system partition, including controllers, backplanes, communication cards, and I/O modules, to be removed without affecting communication with other partitions.
3. Network shall be designed to facilitate speed of communication between networked components and shall not introduce significant data latency.
4. Network shall be designed to have 50 percent spare base system capacity without exceeding performance limits.
5. Provide optical fiber cable, in accordance with Section 26 05 10 – Conductors and Cables, for connections between Network Interface Panel and long-haul fiber optical transmission system.

B. Provide hardware, software, and design to provide fully functional bi-directional data links to other systems, components, PLCs, PACs, and networks as required.

C. Control System shall provide support for the following minimum communication protocols:

1. DNP 3.0 in accordance with IEEE 1815.
2. Modbus Serial (RTU and ASCII).
3. Modbus Network (TCP).
4. HART Communications Protocol.

D. Provide capability for continual capture of NetFlow data between components for network and protocol analysis.

E. Control System Network Switch:

1. Provide Control System network switch.
2. Network Features: Layer 2 Ethernet switching.
3. All network switches shall provide the following minimum features:
    a. Supports linear, star, and ring topology.
    b. Loop Prevention.
    c. Resiliency.
    d. Segmentation.
    e. Prioritization.

f. Time synchronization.

g. Security and Diagnostics.

h. Provide ports and connections, as required. Spare to include, at a minimum, the following: 20 percent spare ports of each type used at the switch.

4. Physical Features:

a. Copper ports: Eight 10/100BaseTX, RJ-45, minimum.

b. Fiber ports: Two 100BaseFX, single-mode 1310nm, ST connectors, minimum.

c. Status indicators on each port.

d. Power: Redundant inputs.

e. Enclosure: Metal case.

5. Network Features:

a. IEEE 802.3 compliance.

b. Full/Half Duplex operation on all ports.

c. 100-Mbps throughput, minimum.

6. Allows enabling/disabling ports.

2.10 CONTROL EQUIPMENT ENCLOSURE

A. Location:

1. As required for mounting system equipment and components.

2. A cabinet shall be located in each Turnout Building and, at a minimum, utilized for mounting the system OIT and related components.

3. Provide adequate room for at least 90 degree opening of cabinet door.

4. No less than 3- by 3-feet accessibility in front of cabinet.

B. Enclosures design shall allow free and easy access to internal components.

C. Type:

1. Wall mounted.

2. NEMA 250 Type 12 enclosure for indoor installations and NEMA 250 Type 4X enclosures for outdoor installations.

3. Size: As required to house components. All new enclosures shall include a minimum of 25 percent spare space.

4. Constructed of sheet metal no lighter than 14-gauge.

5. Seams continuously welded and ground smooth; no holes or knockouts.

6. Continuous hinge on door. Doors shall be lockable.
7. Finish: Manufacturer's standard coating.
8. Back panel.

D. Enclosures shall have louvers, or ventilation, as required to maintain temperature of equipment within manufacturer recommended range. Interior temperature shall be maintained at a minimum of at least 10 degrees F below maximum operating temperature of internal devices at maximum ambient temperature:

1. Fans or other forced ventilation devices shall be provided as redundant pairs and shall be provided with temperature alarms.
2. Louvers and openings shall be equipped with screens and changeable filter material to prevent the entrance of dust, insects, and rodents.

E. Provide thermostatically controlled cabinet space heaters to control condensation in cabinets not in a temperature-controlled environment, or subject to condensation.

F. Provide 120 VAC LED light fixture sufficient to illuminate panel interior for each new enclosure installed by Contractor.

2.11 NETWORK RACK

A. Belden XNM Series as manufactured by Belden Inc., 1N. Brentwood Blvd., St. Louis, MO 63105, or equal having the following essential characteristics:

1. 19-inch standard size.
2. 12U minimum.
3. Lockable.
4. Mesh cooling design.
5. Built-in utility outlets to plug into UPS.
6. Contractor to determine final size based on components to be installed for System Control and Monitoring equipment and Section 28 10 01 – Security and Surveillance System equipment.

2.12 INSTRUMENTS

A. RTD Type Ambient Temperature Sensors:

1. Dwyer Series O-4, or equal, having the following essential characteristics:
   a. Platinum type.
   b. Ambient temperature sensor.
   c. NEMA 4X enclosure.
2. Provide RTD type ambient temperature sensors to monitor ambient temperature inside Turnout Building.

2.13 POWER

A. Provide power distribution system to supplied components, including protection, to support full operation of the system. This distribution system shall include analog loop power supplies and contact interrogation voltages for dry contact inputs where required.

B. Failure of power supply, or power feed, shall be alarmed in the system.

C. Provide fuses, circuit breakers, power supplies, transformers etc., required to supply and transform power feeds at the desired voltage. Provide DIN Rail disconnect for each 120-volt circuit entering enclosure.

D. DC Power Supply:

1. Type:
    a. Switch-mode power supply.
    b. DIN-Rail mountable.
2. Input Voltage: 100-240 VAC.
3. Output Voltage: 24-VDC plus-and-minus 1 percent.
4. Output Current: 5-Amps, minimum.
5. Connection Type: Screw clamp.
6. Operating Temperature Range: -20 to 60 degrees C.

E. Uninterruptable Power Supply (UPS):

1. Type: Tower.
2. Minimum status display and outputs to Control System:
    a. On line.
    b. On battery.
    c. Overload.
    d. Replace battery.
    e. UPS trouble.
3. Input voltage: 120-volts, 60-hertz.
4. Output voltage: 120-volts, 60-hertz.
5. Batteries:
    a. Leak-proof.
    b. Hot-swappable, user replaceable.
    c. Maintenance-free sealed Lead-Acid battery with suspended electrolyte.

6. Size to provide at least 200 percent of full load current for the following equipment inside the Control System cabinet, the Network Interface Panel, and the Communication and Security Rack for at least 30-minutes:
   a. Everything inside Control System Enclosure, Network Interface Panel, and Security and Communication Rack.
   b. Networking and Control System equipment outside of panels and rack.
   c. Fiber optic equipment.
   d. Security and Surveillance Equipment in accordance with Section 28 10 01 – Security and Surveillance System.
   e. Pilot valve solenoid for the Flow Control Valve referenced in Section – 35 22 14 – Valves and Equipment.

7. Receptacle strip(s):
   a. Provide adequate receptacles for each enclosure.
   b. Voltage: 120-volts, 60-hertz.
   c. Current: 20-amperes.
   d. UL listed.

F. Fuse Block and Fuses:
   1. Type: Finger-Safe terminal block.
   2. DIN-Rail mountable.
   3. Fuse:
      a. Rated for size required.
      b. 250-volts, current limiting.

G. Ground Bus:
   1. Material: Copper.
   2. Size: Sufficient to attach equipment ground conductors.
   3. Length: Sufficient to attach equipment ground conductors plus 20 percent spare.
   4. Insulated mounting bracket.

2.14 INSULATED CONDUCTORS

A. Insulated conductors used to connect components external to enclosure are in accordance with Section 26 05 10 – Conductors and Cables.

2.15 SURGE SUPPRESSION

A. 4 to -20 mA Transient Protection Modules:

1. Triple stage hybrid protection.
2. Rated Voltage: 24-VDC
3. Rated Current: 10-Amps
4. Surge Current (8/20 microsecond wave form): 2.5kA, minimum.
5. Clamping Voltage: 30-VDC, maximum.
6. Response Time: Less than 5-nanoseconds.
7. Throughput Resistance: 12-Ohms, maximum.
8. Suppressor condition status indicator.
9. Automatically resets after each transient.

B. Line Voltage Surge Protection Device:

1. Rated Voltage: 120-volt, 60-hertz.
2. Rated Current: 15-Amps.
3. Single-phase service.
4. Surge Current (8/20 microsecond wave form): 7 kA, minimum.
5. Clamping Voltage: 270-volts, maximum.
6. Maximum Continuous Operating VAC (MCOV): 115 percent rated line voltage.
7. Response Time: Less than 5-nanoseconds.
8. Modes of Protection: L-N, L-G, N-G.
9. Suppressor condition status indicator
10. Automatically resets after each transient.

C. For automation equipment in outdoor enclosures, surge suppression is required for power supply inputs and wiring leaving enclosure, including digital I/O, analog I/O, and communication wiring.

2.16 COMPONENT LIFE CYCLE / SUPPORT

A. Components for Control System shall be chosen to maximize installation life-cycle and extend the time to replacement to the extent possible. Components shall be chosen with the following characteristics to help provide maximum life-cycle of the installation:

1. Components from established product lines and manufacturers.
2. Components from the most recent generation of equipment available from their manufacturer. Components where a later generation of equipment with a similar functionality is available shall not be acceptable except where specifically identified in this Section.

B. Software packages used in the installation and for development of system graphics and control logic, shall be up-to-date and compatible with the most recent version of major operating systems, including Microsoft Windows.

2.17 SOURCE QUALITY CONTROL

A. Government reserves the right to inspect equipment at supplier's manufacturing plant prior to final testing.

PART 3 EXECUTION

3.01 INSTALLATION

A. In accordance with manufacturer's recommendations, with suitable and secure fasteners.

B. Incorporate into the Control System, at a minimum, the instrumentation and control I/O points identified in Table 25 00 01A – Typical Turnout Building I/O and Data Point List and the P&IDs shown in drawings 1695-D-62569 through 1695-D-62577.

C. For connections to the programmable controllers which run outside the Turnout Building, surge suppression is required, including digital and analog I/O.

D. Coordinate with Government for the installation of the Network Interface Panel, which is installed by Others in Turnout Building.

1. Do not make any modifications or alterations to the Network Interface Panel.

E. The contractor shall furnish and install optical fiber in the Turnout Building site per Section 26 05 10 – Conductors and Cables.

F. Contractor shall provide fiber connection between long haul fiber communication system splice box to the Turnout Building and make fiber installation ready for others to complete installation to the Network Interface Panel. Utilize fiber splice handholes shown on supplied drawings to route fiber to Turnout Building.

G. Contractor is responsible for determining the final installed layout, wiring connections and schematic diagrams based on field installed equipment and design coordination meetings.

H. Make wire connections between the Control System equipment cabinet terminal blocks and the programmable controller's field wiring arms with NEC type SIS No. 18 AWG switchboard wire.

I. Make connections necessary to place the Control System into operation.

J. Terminal Blocks:

1. Arrange terminal blocks to segregate signal types and account for spare space.

2. Arrange terminal blocks to allow external cabling to be supported.

3. Mount terminal blocks at least 3-inches from panel edges and other devices. Mount adjacent rows of terminal blocks with 12-inches separation minimum. Access to front of terminal blocks shall be unobstructed.

4. Machine lettered terminal numbers on terminal block marking strips. Do not use hand lettering.

5. Identify vertical columns of terminal blocks with single letter such as "A", "B", or "C". Vertical columns may contain more than 12 termination points.

6. Terminals:

   a. Binding-head or washer-head screws with serrated or grooved contact surfaces.

   b. Spring-clamp type terminal blocks are not acceptable.

7. Voltage rating: 300-volts.

8. Current rating: 20-amps.

9. Analog inputs shall be disconnecting switch type terminal blocks.

K. Cable and Wiring:

1. Provide cable and wiring in accordance with Section 26 05 10 – Conductors and Cables and Section 26 05 02 – Basic Electrical Materials and Methods.

2. Wire Connections:

   a. Make connections at device terminals or terminal blocks. Maximum two wires per termination point.

   b. Install incoming 120-volt branch circuits on adjacent terminal block terminals.

   c. Install ring tongue connectors for terminations at terminal blocks and electrical devices. Use pin-type terminal connectors where use of ring tongues is not practical such as terminations at selector switches, pushbuttons, indicating lamps, and auxiliary relays.

   d. Do not terminate wire without terminal connector.

   e. Wire splices are not acceptable.

L. Line power for control and monitoring equipment, including Network Interface Panel, shall be supplied from Contractor provided UPS system.

M. Network cabling shall provide a designated and easily identified to-and-from location marker at each end of cable.

N. Make cable and conduit connections. Use watertight conduit hubs for conduit connections.

O. Make grounding and bonding connections in accordance with Section 26 05 20 – Grounding and Bonding.

P. Control Cable and Wire Identification: In accordance with Section 26 05 10 – Conductors and Cables.

Q. Provide software and associated programming/configuration required to meet performance requirements of Contract Documents:

1. At substantial completion of Project, turn current licenses for software over to Government in Bureau of Reclamation's name and install latest version, upgrade or service pack for all software.

2. Provide respective software supplier's Comprehensive Support Contract for software and network components covering full one-year warranty period following substantial completion. Provide software upgrades, service packs and tech support from software or network component suppliers at no cost.

R. Nameplates: Provide nameplates for Contractor supplied equipment in accordance with Section 26 05 02 – Basic Electrical Materials and Methods.

S. Installation shall look complete and professionally done, damage to enclosures or installation area shall be repaired and left in a finished condition.

T. I/O and Remote Interface Data Lists:

1. List is solely for information of Contractor. I/O list may change throughout Contractor system design period.

2. I/O Table and Data Point List are shown as applied to Sanostee Turnout Building, but are the same for every Turnout Building, only the location tag number will change. All I/O and Data shown here apply across all Turnout Buildings.

Table 25 00 01A – Typical Turnout Building I/O and Data Point List

Equipment	Description	Type	Signal
S1205-FIT-001	Flowmeter Flowrate	MB	RTU
S1205-FIT-001	Flowmeter Totalizer	MB	RTU
S1205-FIT-001	Flowmeter Direction	MB	RTU
S1205-EX-001	UPS Online	MB	RTU
S1205-EX-001	UPS On Battery	MB	RTU
S1205-EX-001	UPS Overload	MB	RTU
S1205-EX-001	UPS Replace Battery	MB	RTU
S1205-EX-001	UPS Trouble	MB	RTU
S1205-FCV-001	Flow Control Valve Close Command	DO	24 VDC

S1205-ZSC-001	Flow Control Valve Closed	DI	24 VDC
S1205-ZSC-001	Flow Control Valve Not Closed	DI	24 VDC
S1205-TE-001	Indoor Ambient Temperature	AI	4-20 mA
S1205-ZSC-002	Control System Enclosure Intrusion	DI	24 VDC
S1205-ZSC-003	Network Interface Panel	DI	24 VDC
S1205-ZSC-004	Comm. & Security Rack Intrusion	DI	24 VDC
S1205-ZSC-005	Site Entrance Gate Intrusion	DI	24 VDC
S1205-ZSC-006	Building Door 1 Entrance Intrusion	DI	24 VDC
S1205-ZSC-007	Building Door 2 Entrance Intrusion	DI	24 VDC
S1205-ZSC-008	Fiber Splice Vault Intrusion	DI	24 VDC

Table 25 00 01B – SJLWTP SCADA System Remote Interface Data Point List

Equipment	Description	Type	Signal
S1205-FIT-001	Flowmeter Flowrate	Output	MB TCP
S1205-FIT-001	Flowmeter Totalizer	Output	MB TCP
S1205-FIT-001	Flowmeter Direction	Output	MB TCP
S1205-EX-001	UPS Trouble	Output	MB TCP
S1205-FCV-001	Flow Control Valve Close Command	Input	MB TCP
S1205-ZSC-001	Flow Control Valve Closed	Output	MB TCP
S1205-ZSC-001	Flow Control Valve Not Closed	Output	MB TCP
S1205-TE-001	Indoor Ambient Temperature	Output	MB TCP
Control System	General Intrusion Alarm	Output	MB TCP
Control System	Control System Trouble	Output	MB TCP
Control System	In Remote	Output	MB TCP

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Perform field wiring checkout and test in accordance with Section 26 05 90 – Wiring Checkout and Tests.

B. Coordinate with Government to perform commissioning and Lateral Integration Testing in accordance with Section 01 80 15 – Commissioning.

END OF SECTION

SECTION 33 11 12
STEEL LINE PIPE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Line Pipe for BBN9 Turnout:

1. Payment: Lump sum price offered in Price Schedule 4.

a. Includes pipes, valves, and associated appurtenances shown on Drawings 1695-D-62548.

b. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and placing materials, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

B. Line Pipe for Coyote Canyon:

1. Payment: Lump sum price offered in Price Schedule 5.

a. Includes cost of the following:

1) Pipe trench excavation in accordance with Section 31 23 22 – Pipe Trench Earthwork.

2) Furnishing and installing Coyote Canyon Turnout as shown on Drawing 1695-D-62550, including but not limited to the following:

a) Fittings, bedding, embedment, utility markers, concrete, reinforcing steel, blocking and collars, not included elsewhere.

b) Laying pipe.

C. Does not include pipe specified in Section 35 21 95 – Steel Manifold Piping.

1.02 PROJECT CONDITIONS

A. Refer to Drawing 1695-D-62548, 1695-D-62549, 1695-D-62550, 1695-D-62551, 1695-D-62552, 1695-D-62553, and 1695-D-62554.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM)

1.	ASTM A139/A139M-16	Electric-Fusion (Arc) - Welded Steel Pipe (NPS 4 and Over)
2.	ASTM C150/C150M-17	Portland Cement
3.	ASTM A283/A283M-18	Low and Intermediate Tensile Strength Carbon Steel Plates
4.	ASTM A1011/A1011M-15	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Allow and High-Strength Low-Allow with Improved Formability
5.	ASTM A1018/A1018M-16a5	Steel, Sheet and Strip, Heavy Thickness Coils, Hot Rolled, Carbon, Structural, High-Strength Low-Alloy, Columbium or Vanadium, and High-Strength Low-Alloy with Improved Formability
6.	ASTM E165/E165M-12	Liquid Penetrant Examination for General Industry
7.	ASTM E709/E709M-15	Standard Guide for Magnetic Particle Testing

B. American Welding Society, Inc. (AWS)

1.	AWS D1.1/D1.1M-15	Structural Welding Code – Steel

C. American Water Works Association (AWWA)

1.	AWWA C200-12	Steel Water Pipe - 6 Inch (150mm) and Larger
2.	AWWA C205-12	Cement-Mortar Protective Lining and Coating for Steel Water Pipe - 4 Inch (100mm) and Larger - Shop Applied
3.	AWWA C206-17	Field Welding of Steel Water Pipe
4.	AWWA C208-12	Dimensions for Fabricated Steel Water Pipe Fittings
5.	AWWA C227-11	Bolted, Split-Sleeve Restrained and Non-restrained Couplings for Plain-End Pipe
6.	AWWA C604-11	Installation of Buried Steel Water Pipe 4 Inch (100 mm) and Larger

7.	AWWA M11-04	Steel Pipe: A Guide for Design and Installation, Fourth Edition

1.04 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals.

B. RSN 33 11 12-1, Shop Drawings:

1. Show pipe and fitting fabrication details.
2. Show exact dimensions of joints, position, marking of pipe sections and proposed joint restraint.
3. Show exact dimensions of welding lead holes and repair method.
4. Show details of factory epoxy lining of joint cement mortar hold back area if alternate epoxy joint recess lining is proposed.
5. Include centerline and invert stationing and elevations at horizontal and vertical changes in alignment, and subgrade elevation for each pipe segment and fitting.
6. Include pipe thickness or pressure class of pipe segment or fitting.

C. RSN 33 11 12-2, Ventilation Plan:

1. Method for ventilating inside of pipe when placing mortar lining for joints.

D. RSN 33 11 12-3, Field Welder Qualifications

1. Welder certifications

E. RSN 33 11 12-4, Filling and Testing Plan:

1. Proposed rate, time, and procedure for:
   a. Cleaning.
   b. Filling.
   c. Field and pressure testing.
   d. Draining pipeline, if not included in RSN 33 11 12-7.
2. Method for disposing of water drained from pipeline to enable repair of leaks.
3. Leak testing equipment and data sampling plan.

F. RSN 33 11 12-5, Flotation Prevention Plan:

1. Show location and describe method of preventing pipe from floating.
2. Provide supporting calculations.

G. ***RSN 33 11*** 12-6, Pipe Deflection Measurements.

H. RSN 33 11 12-7, San Juan Lateral Outage Plan.

1. Provide detailed outage plan covering outages required on the San Juan Lateral.

a. Show compliance with 01 14 12 – Work Restrictions and Limitations.

b. Include required permits for discharge of water.

c. Include a schedule, isolation limits, discharge amount, pumping requirements, equipment, discharge locations, required coordination and Government support.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Ship pipe after receiving COR approval as per Section 2.03 – Source Quality Assurance.

B. Prevent damage to pipe and fittings during loading, transporting, unloading, storing, and laying.

C. Transport metallic pipe and fittings on padded bolsters curved to fit outside of pipes. Use heavy padding under ties.

D. Tightly close open ends of shop-applied, cement-mortar-lined pipe with plastic wrap for protection of cement-mortar lining during shipment:

1. Plastic Wrap:

a. At least 2 thicknesses of 6-mil sheet polyethylene plastic.

b. Remain on pipe until installation.

E. Support and store pipe above ground surface. Do not allow bells and spigots to contact each other or the ground.

F. Government will inspect pipe once it is delivered.

G. Replace or repair, as approved by COR, any pipe that is damaged during shipment, storage, or installation at Contractor's expense. Include linings and coatings.

1.06 QUALIFICATIONS

A. Pipe manufacturers shall be certified in accordance with ISO:9001 or similar nationally certified program approved by COR.

B. Line pipe and specials manufacturers:

1. 10-years successful experience producing products as specified.

2. Provide references for at least 3 completed projects with similar pipe diameters (within 12-inches) and pressures.

C. Pipe Layers: Skilled and experienced in laying pipe with joints as needed for each material type of pipe used. Minimum 2-years installing pipe with similar pipe diameters (within 12-inches), pipe material, and pressures:

1. In the absence of specific pipe material experience, substitution of onsite pipe manufacturer training and quality control is allowed:
    a. Provide manufacturer's representative certified in pipeline installation, including all applicable joints, for training pipe layers and pipe foreman prior to installation.
    b. In lieu of specific pipe material installation experience, pipe layers and pipe foreman shall still have experience with installing similar pipe diameters and pressures.

D. Qualify welding procedures and welders in accordance with code under which welding is specified.

PART 2 PRODUCTS

2.01 STEEL LINE PIPE

A. Pipe:
1. Electric fusion (arc) welded helical-seam steel pipe: ASTM A139, Grade C, D, or E.
2. Fabricated in accordance with AWWA C200:
    a. Except:
        1) Steel plate: ASTM A283, Grade C or D.
        2) Steel Sheet or coil: ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Inside diameter measured to inside of lining shall not be less than nominal diameter shown on drawings.

C. Minimum Steel Wall Thickness:
1. Greater thickness from the following requirements:
    a. Steel Wall Thickness for Handling:
        1) Equal to inside diameter (inches) of pipe steel shell divided by 240 for mortar lined and flexible coated steel pipe.
    b. Steel Wall Thickness for Internal Pressure:
        1) Hoop stress of pipe shell. Not to exceed:
            a) 50 percent yield strength as defined by steel grade of steel using the Barlow formula. With pressure defined by Hydraulic Grade Line as shown on pipeline plan and profile drawings.

b) 75 percent yield strength as defined by grade of steel using the Barlow formula. With pressure defined by pipe head class as shown on pipeline plan and profile drawings.

2) Barlow formula as defined by AWWA M11 Equation 4-1.

D. Steel shall be fully killed and conform to fine grain practice.

E. Rubber gaskets: In accordance with AWWA C200.

F. Materials and chemicals that may come into contact with drinking water shall be certified by NSF 61.

2.02 FITTINGS

A. Steel: ASTM A283, Grade C or D or ASTM A1011, designation SS, Grade 40, 45, or 50; or ASTM A1018, designation SS, Grade 40 or designation HSLAS-F, Grade 50.

B. Minimum steel wall thickness and fitting dimensions: In accordance with AWWA C200, AWWA C208 and AWWA M-11.

C. Welding: In accordance with AWS D1.1.

D. Lifting eyes and other handling devices: Made part of fitting before lining and coating are applied.

E. Bolts and Nuts: In accordance with ASME B18.2.1 and FS FF-N-836.

F. Coating and lining in accordance with Section 09 96 20 – Coatings.

G. Cement-Mortar Lining for Specials and Bends Larger than 24-inches:

1. In accordance with AWWA C205.
2. Apply lining with a resultant surface finish, including joints, equal to finish of adjacent pipe.

H. Joints between fittings and ductile iron pipe:

1. Rubber gasket joints.
2. Joint dimensions and tolerances: Same as pipe manufacturer's joint design.

I. Miter Bends: Fabricate steel bends in accordance with AWWA C208, AWWA M-11 and as shown on drawings.

J. Pipe Couplings:

1. Coupling types as shown on drawings.
2. Suitable for line pipe materials used.
3. Suitable for cold water pressures of pipe head classes on drawings.

4. Refer to Section 35 21 95 – Steel Manifold Piping for coupling requirements.

K. Closure Section Joints

1. Restraint requirements shown on drawings.
2. Field welded butt strap joints.
3. Bolted, Split-Sleeve Restrained and Nonrestrained Couplings for Plain – End:
   a. In accordance with AWWA C227.
   b. Refer to Section 35 21 95 – Metal Piping for Line Pipe for coupling requirements.

L. Tees:

1. Tees for air valves, blowoffs, manholes, and temporary construction line as shown on drawings.
2. Tee length, minimum: As shown on drawings, required for blocking, or in accordance with AWWA C208 whichever is greater

2.03 LININGS AND COATINGS

A. In accordance with Section 09 96 20 – Coatings.

B. Do not field apply mortar lining without OGR approval.

2.04 CONCRETE

A. Concrete in Thrust Blocks and Collars: Refer to Section 03 30 00 – Cast-in-Place Concrete.

2.05 FLANGES:

A. In accordance with AWWA C207.

B. Class: Working pressure meets or exceeds head class of attached pipe

2.06 FLANGE GASKETS:

A. BLUE-GARD Style 3000 manufactured by Garlock Sealing Technologies, 1666 Division Street, Palmyra NY 14522, or equal, having the following essential characteristics:

1. Sized in accordance with AWWA C207, Table 1.
2. For AWWA C207 flanged joints.
3. Compressed, Non-Asbestos (CNA) Gasketing with Aramid Fibers and a NGR Binder.
4. For potable cold-water service.

2.07 PIPE ACCESSORIES

A. Sectionalizing Valve, Air Valve, and Blowoff Assemblies: In accordance with Section 35 22 14 – Valves and Equipment.

B. Buried Manhole Assemblies:

1. In accordance with Section 03 48 00 – Precast Concrete Structures.
2. Cement-mortar lining: AWWA C205, in accordance with shop applied mortar for lining of specials, except:
    a. Reinforcement required for pipe larger than 24-inch diameter.
    b. Cement: ASTM C150, Type V.
3. Required at locations shown on drawings.

C. Warning Tape:

1. Polyethylene tape.
2. Thickness: 4 mils.
3. Magnetically detectable.
4. Width, minimum: 6-inches.
5. Color: Blue with silver lettering.
6. Legend repeated at least every 3-feet: "CAUTION: WATERLINE BURIED BELOW - FOR MORE INFORMATION CONTACT THE NAVAJO TRIBAL UTILITY AUTHORITY".

2.08 CONTRACTOR SOURCE QUALITY TESTING

A. Pipe and Fittings:

1. Hydrostatic Test:
    a. Perform shop hydrostatic test on pipe which stresses steel to 75 percent of minimum yield point of the steel.
    b. Hold pressure long enough to allow thorough inspection of welded joints.
    c. Repair leaks by rewelding and retesting joints.
    d. Test sections prior to forming bell and spigot joints.
    e. Repair defects and retest section before applying lining and coating.
    f. Government may witness hydrostatic testing and calibration of pressure gauges. Notify Government 30 days prior to performing hydrostatic test.
2. Fittings fabricated from tested steel pipe do not require hydrostatic testing if welds are tested.
    a. Weld test: Dye penetrant test in accordance with ASTM E165 or Magnetic Particle in accordance with ASTM E709.

b. Weld defect repair. Chip, flame gouge, or grind to sound metal; reweld; and test repaired weld.

B. Joint Testing: Assemble one joint for each pipe diameter to check bell and spigot to check fit prior to coating or lining.

PART 3 EXECUTION

3.01 GENERAL

A. Perform cleaning, filling, and testing after backfill has been placed to finished grade or as approved by OGR.

B. Responsible Surveyor licensed in State of New Mexico or a certified Construction Surveyor, as defined in and in accordance with Section 01 71 20 – Surveying, shall be onsite during excavation and pipelaying.

C. Protect pipe from contamination, damage, and elements during storage on-site by covering with tarp as directed by OGR.

D. Install pipe based on these specifications or pipe manufacturer's recommendation, whichever is more stringent.

E. Isolate dissimilar metals.

3.02 INSTALLATION

A. Keep openings to installed pipe closed with watertight inflatable plugs during work stoppage, including end of work day, breaks, and work delays.

B. If pipe is flooded during construction, clear floodwater by draining and flushing with water, or other approved method, until pipe is clean.

C. Lubricant:

1. Keep clean.
2. Apply with dedicated, clean applicator brushes.
3. Apply lubricant as specified by pipe manufacturer:
    a. Use only lubricant supplied by pipe manufacturer.
    b. Lubricate spigot end of pipe as recommended by manufacturer.
    c. Do not lubricate either gasket or gasket groove in bells where gaskets are field installed.
    d. Lubricate factory-installed, non-removable-type gaskets as recommended by pipe manufacturer.

D. Coat buried manholes with cement mortar as shown on drawings.

E. Install warning tape over center of pipe at least 18-inches below ground and at least 18-inches above pipe.

3.03 LAYING PIPE

A. Excavate pipe trench in accordance with Section 31 23 22 – Pipe Trench Earthwork.

1. Grade pipe trenches to provide uniform slope along bottom of pipe.
2. At joints involving bells or collars, provide holes at joint of ample size to prevent bells or collars from coming in contact with subgrade and to allow for bedding placement.

B. Keep pipe trenches free of water during pipelaying operations.

C. Lower pipe into trench and place pipe in position such that no soil gets inside pipe and pipe is not damaged.

D. Install pipe to line and grade as shown on drawings.

E. On grades exceeding 10 percent, lay pipe uphill.

F. When pipelaying is not in progress, keep ends of pipelines closed with watertight inflatable plugs.

G. Joining Pipe:

1. In accordance with AWWA C604.
2. Verify inside joint gap meets manufacturer's tolerances as work progresses.

H. Joints for Pipe with Welded Joints:

1. In accordance with AWWA C206. Welded joints after backfill not allowed.
2. Perform dye penetrant test on welds in accordance with ASTM E165.
3. Welded Butt Joints:
    a. As required for pipe on steep slopes as shown on drawings.
4. Double Lap Weld Joints:
    a. Provide joint with full fillet welds.
    b. Air test double lap welds in accordance with AWWA C206.
    c. Required on thrust restrained joints greater than 30 degree bend deflection angles, unless butt welded joints are required.
5. Single Lap Weld Joints:
    a. May be used on thrust restrained joints with 30 degree bend deflection angles or less.
    b. Road and wash crossings unless butt or double-lap welded joints are required.

I. Connections at Thrust Blocks and Structures: As shown on drawings

J. Closure Sections:

1. Where necessary as determined by Contractor and subject to approval of OGR.
2. Ambient temperature when closure section is welded, maximum: 60 degrees F.
3. Assemble push-on or mechanical joints in accordance with applicable AWWA standard for pipe type and manufacturer's recommendations.
4. If adjustment of the position of length of pipe is required after installation, remove and relay length of pipe as new pipe.
5. Maintain pipe firmly in final position.
6. Placing Elastomeric Gasket:
   a. Clean gasket, bell, especially the groove, and spigot with rag, brush, or paper towel to remove any dirt or foreign material.
   b. Use only gaskets which are designed for and supplied with pipe.
   c. Lubricate pipe as recommended by manufacturer.
   d. Spigot Groove Method:
      1) Follow manufacturer's recommended practices for gasket installation.
      2) After placing elastomeric gasket in spigot groove, equalize elastomeric gasket cross section by inserting tool such as a large screwdriver under elastomeric gasket and moving it around periphery of pipe spigot.
   e. Coupling Groove Method: Follow manufacturer's recommended practices for gasket installation. Typically, factory installed.
7. Fit pipe unit's together, spigot to bell or coupling, and draw joints together so that bells or couplings and spigots are fully engaged with uniform contact to gasket.
8. Do not swing or "stab" joint and do not suspend pipe and swing into bell or coupling.
9. Fit pipe units together in a manner to avoid twisting or otherwise displacing or damaging elastomeric gasket.
10. After joining pipe units, insert feeler gauge between pipe bell and pipe spigot, move it around periphery of pipe to determine that position of elastomeric gasket is correct and there are no fish-mouth problems. If adjustment of position of a length of pipe is required after installation, remove and relay length of pipe as new pipe.

K. Changes in Alignment and Grade:

1. Make changes in alignment and grade with miter bends as shown on drawings.
2. Make other minor changes in alignment and grade by providing small deflections between adjacent pipe:
   a. Do not exceed manufacturer's deflection tolerances.

b. Pipelaying diagrams shall only deflect joints no more than 50-percent of manufacturer's maximum deflection tolerances.

3. Provide restrained joint lengths upstream and downstream of bends or thrust blocks as shown on drawings. Ductile iron fittings with restrained joint lengths as limited by drawings may be used if Contractor submits revised plan and profile drawings for approval.

4. Do not encase rubber gasket joints in concrete.

5. Refer to individual pipe Sections for additional requirements. If requirements conflict, use most stringent requirement.

L. After pipelaying and joining operations are completed, clean inside of pipe and remove debris.

M. Schedule work so that at no time pipe remains in trench more than 7-days before backfill is placed to original ground surface or to other specified backfill limits shown on drawings.

3.04 THRUST RESTRAINT

A. Restrained joint couplings or concrete blocking and collars as shown on drawings. Refer to Section 03 30 00 – Cast-in-Place Concrete and in accordance with applicable sections listed in 2.01.A.

B. Plan, profile, and restraint system shown on drawings require modification if alternate methods are used.

C. Alternate methods of restraint may not be used unless submitted and approved by COR.

3.05 TOLERANCES

A. Lay pipe to lines and grades shown on drawings or established by OGR to following tolerances:

1. Total departure from established alignment and grade, maximum: 1-inch.

3.06 ADJUSTMENTS IN ALIGNMENT AND GRADE FOR PULLED JOINTS

A. Per approved shop drawings.

B. Make small minor changes in alignment and grade by providing small deflections between adjacent pipe sections as follows:

1. Pulling rubber gasket bell-and-spigot joint:

a. A total 1-inch joint deflection may be permitted by reducing normal 1/2-inch inside mortar space to 1/4-inch and pulling opposite side of pipe 3/4-inch from normal closure.

b. Provide unsymmetrical closure at welded lap joint up to maximum pullout of 1-inch.

2. Maximum deflection angle between adjacent pipe sections: Manufacture's recommendations but shall not exceed 5 degrees.

C. Lay ends of each section of steel line pipe on theoretical centerline of curve and to grade shown on drawings within laying tolerances prescribed above.

3.07 JOINT LINING AND COATING

A. Field joints for pipe with shop applied cement-mortar lining.

1. Line field joints with cement-mortar in accordance with AWWA C205.
2. Coat exterior joints as specified in Section 09 96 20 – Coatings.
3. Do not backfill field joints until coating is approved by OGR.

B. Apply shrink sleeves to field joints in accordance with Section 09 96 20 – Coatings.

3.08 BACKFILL

A. Keep internal supports in place until embedment has been placed and compacted above bottom of pipe to minimum height of 0.7 times diameter.

3.09 *FILLING AND DRAINING PIPELINE*

A. Notify COR 4 days prior to filling and testing.

B. Water for filling and testing pipeline in accordance with Section 01 51 00 – Temporary Utilities

C. San Juan Lateral Pipeline

1. Fill rate, maximum: 2 cubic feet per second

2. Drain rate, maximum: 3 cubic feet per second.

D. Dispose of water in accordance with permits in Section 01 57 30 – Water Pollution Control.

3.10 PIPE DIAMETER DEFLECTION

A. Allowable short-term vertical pipe diameter deflection after backfilling is complete.

1. Decrease, maximum: 2 percent of nominal pipe diameter.
2. Elongation, maximum: 3 percent of nominal pipe diameter as measured when backfill reaches pipe crown.
3. Short term is within 2 weeks after backfilling is complete.

B. Deleted

END OF SECTION

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SECTION 33 13 00
DISINFECTION OF WATER UTILITY DISTRIBUTION

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Disinfection of Water Utility Distribution:

1. Payment: Lump sum price offered in Price Schedules.
    a. Include the cost of ***disinfecting pipes***.

1.02 REFERENCE STANDARDS

A. American Water Works Association (AWWA):

1.	AWWA B300-24	Hypochlorites
2.	AWWA C604-17	Installation of Buried Steel Water Pipe-4 In. (100 mm) and Larger
3.	AWWA C651-23	Disinfecting Water Mains
4.	AWWA C655-25	Field Dechlorination

B. National Sanitation Foundation International / American National Standards Institute / National Standard of Canada (NSF/ANSI/CAN)

1.	NSF/ANSI/CAN 60-2025	Drinking Water Treatment Chemicals – Health Effects
2.	NSF/ANSI/CAN 61-2025	Drinking Water System Components – Health Effects

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 – Submittals:

B. RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan:

1. Demonstrate compliance with AWWA C604, AWWA C651 and AWWA C655.
2. Show product compliance with NSF/ANSI/CAN 60 and/or NSF/ANSI/CAN 61.
3. Permits and approvals:
    a. Identify required permits and approvals.
    b. Plan, procedures, and schedule.
4. Initial Flushing:
    a. Plan, procedures, and schedule.

b. Include flushing locations, water source, water volume, and flow rate.

5. Hydrostatic Pressure and Leakage Testing:

   a. Plan, procedures, and schedule.

   b. Include pipe segment materials, pipe diameters, pipe stations, water source, water volumes, pressure gauge measurement locations, test pressure calculations, acceptable makeup water volume calculations, and pass/fail determinations.

6. Disinfection:

   a. Plan, procedures, and schedule.

   b. Include disinfection method, disinfectant type and form, disinfectant dose, chlorine residual sample locations, disinfectant storage, disinfectant disposal, water source, water volume, and flow rate.

   c. If disinfection will be performed in multiple segments, identify and delineate each segment.

7. Final Flushing:

   a. Plan, procedures, and schedule.

   b. Include flushing locations, water source, water volume, and flow rate.

   c. Include dechlorination plan and procedures.

8. Verification:

   a. Plan, procedures, and schedule.

   b. Include proposed sampling locations.

   c. Identify independent testing laboratory, including certifications/qualifications to perform bacteriological testing.

9. Plan and procedures for prevention of recontamination of system components/segments once they are disinfected.

10. Safety:

    a. Chemical spill response plan for any chemicals to be used.

    b. Job Hazard Analysis (JHA), including, but not limited to:

       1) Personal protective equipment required for handling disinfection and dechlorination chemicals.

       2) Emergency eyewash and shower locations.

       3) Local emergency responder communication information.

C. RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report:

1. Detailed report demonstrating flushing and disinfection activities were accomplished in accordance with AWWA C651 and AWWA C655.

2. Examination
    a. Name of field person in charge of examination.
    b. Date and time of examination.
    c. Include verification of existing conditions.
    d. Include verification of pipe and associated equipment cleaning.
    e. Include verification of equipment operations.
3. Initial Flushing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completion.
    c. Flow rate and velocity achieved.
    d. Any data collected or readings taken during flushing activities.
4. Hydrostatic Pressure and Leakage Testing:
    a. Name of field person in charge of initial flushing.
    b. Date and time for start and completed.
    c. Include initial and final pressure readings for each measurement location, test pressures achieved, makeup water volumes required, pressure gauge serial numbers and calibration certification dates, duration of conditioning period and formal test periods, and signatures of responsible parties certifying the testing results.
    d. Test records for each test performed, including failures.
5. Disinfection:
    a. Name of field person in charge of disinfection and sampling.
    b. Type and form of disinfectant used.
    c. Date and time for disinfection start and completion.
    d. Chlorine residual sample location names identified on drawing or map.
    e. Table with the following for each chlorine residual sample location:
        1) Location name.
        2) Date and time of samples taken.
        3) Free chlorine residual concentration at each sample time.
    f. Any other data collected or readings taken during disinfection to verify compliance with AWWA C651 and AWWA C655.
6. Final Flushing:
    a. Name of field person in charge of final flushing.
    b. Date and time for start and completion.

c. Chlorine residual sample locations and discharge location identified on a drawing or map.

d. Any data collected or readings taken during final flushing activities, including, but not limited to, free chlorine concentrations at identified sample locations.

7. Bacteriological Test Results:

a. Bacteriological sample location names identified on drawing or map.

b. Table with the following for each bacteriological sample location:

1) Location name.

2) Name of sample collector.

3) Date and time of each sample.

4) Free chlorine residual test results.

5) Total coliform bacteria test results.

6) Escherichia coli bacteria test results.

7) Heterotrophic plate count (HPC) bacteria test results.

8) Certify water meets or does not meet bacterial test requirements.

c. Provide data summary and necessary graphs and plots of tests performed.

d. Testing laboratory: name, address, and telephone number.

e. Testing laboratory report(s) attached, including sample chain(s) of custody.

f. Water Quality Certificate: Certified, signed, dated and attested to having met the requirements of AWWA C651.

8. Issues and Resolutions Log:

a. Include any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan, as approved by the COR.

9. Field Check Prints: Marked field check prints with drawing modifications in accordance with Sections 01 33 26 – Electrical Drawings and Data and 01 78 30 – Project Record Documents.

a. Use in preparing Final Drawings, including, but not limited to, the addition of any taps for sampling.

1.04 QUALIFICATIONS

A. Entities preparing disinfection documentation or performing disinfection and related activities shall have at least 10 years' experience performing similar services.

B. The laboratory performing bacteriological testing to confirm successful disinfection shall be certified by the State of New Mexico to perform the required analyses for drinking water compliance.

PART 2 PRODUCTS

2.01 GENERAL

1. In accordance with NSF/ANSI/CAN 60.
2. In accordance with NSF/ANSI/CAN 61.

PART 3 EXECUTION

3.01 GENERAL

A. In accordance with AWWA C604, AWWA C651 and AWWA C655, including, but not limited to, flushing, testing, disinfection, and bacteriological testing.

1. Optional sections of AWWA C651 that are required:
    a. 5.2.9 Backflow protection.
    b. Bacteriological testing at all sampling locations shall include total coliform, E. coli, and standard heterotrophic plate count (HPC) analyses.

B. In accordance with the approved version of RSN 33 13 00-1, Flushing, Testing, Disinfection, and Verification Plan. Any deviations from the approved Flushing, Testing, Disinfection, and Verification Plan must be approved by the COR.

C. Flushing, testing, disinfection, and verification activities shall be performed after completion and approval of all construction testing in accordance with specifications.

D. Provide water, chemicals, equipment, instrumentation, testing, and personnel necessary for performing flushing, testing, disinfection, and verification activities.

E. Potable water shall be used for all flushing, testing, disinfection, and verification activities.

F. Once disinfected, pipeline, booster pump stations, and appurtenances shall remain isolated and not be exposed to any potential or actual recontamination. A newly disinfected portion of the system shall only be connected to existing or previously disinfected portions of the system once disinfection is confirmed via bacteriological testing and written permission is obtained from COR.

3.02 COORDINATION

A. Coordinate work activities in accordance with Section 01 31 19 – Project Management and Coordination.

B. Government may witness any flushing, testing, disinfection, and verification activities.

C. Contractor shall:

1. Not schedule flushing, testing, disinfection, and verification activities until Flushing, Testing, Disinfection, and Verification Plan is approved.
2. Notify COR in writing a minimum of 60 days prior to beginning any flushing, testing, disinfection, and verification activities.
3. Complete all construction testing and request written confirmation of completion prior to beginning flushing, testing, disinfection, and verification activities.
4. Notify COR in writing of scheduled activities a minimum of 14 days prior to performing activity to allow Government opportunity to witness.
5. Coordinate with COR, NTUA, and water distribution system operators prior to commencing flushing and disinfection and during flushing and disinfection.
6. Ensure work is coordinated with commissioning, startup and ongoing operations activities along with ongoing distribution system operations.
7. Hold daily coordination meetings one week prior to Initial Flushing with COR, NTUA, and water distribution system operators.

3.03 EXAMINATION

A. Verify existing conditions before starting work.

1. Designated flush water discharge location(s) is acceptable and capable of receiving the volume and flow rate of the flush without causing:
   a. Flooding or erosion at the discharge location(s).
   b. Damage to adjacent infrastructure.
   c. Environmental violation.
2. All connections to the existing live system are closed and confirmed closed.
3. Pressure differentials are calculated and documented to ensure flow in intended direction upon valve openings.
4. No cross-connections exist between the new untested, unflushed system and the potable distribution system.

B. Verify pipe and associated equipment has been cleaned and are operational.

1. Visual inspection of accessible pipe interiors shall be performed, including:
   a. Inspection of all open pipe ends before jointing the connections.
   b. Confirmation that no visible debris, standing water, mud, rocks, tools, rags, or construction materials remain inside the pipe.
   c. Inspection of all appurtenance vaults and valve boxes for debris.

2. All gate and butterfly valves have been operated through full open and close cycles.
3. All air release/vacuum valves are installed, operational, and will function during the disinfection fill to allow air purging.
4. Meters are in place and ready to receive flow.

3.04 INSTALLATION

A. Authority Having Jurisdiction (AHJ) (e.g., NMED):

1. Flushing, disinfection, and verification activities shall be performed in accordance with AHJ requirements.
2. Obtain AHJ approvals and permits before commencing flushing, testing, disinfection, and verification activities.

B. Initial Flushing.

1. Prior to Initial Flushing, complete Examination.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

C. Hydrostatic Pressure Test and Leakage Test.

1. In accordance with AWWA C651, AWWA C604, and the AHJ.

   In the event of conflicting standards, the more stringent requirement governs the testing procedures.
2. Appurtenances to be included in the hydrostatic pressure test.
3. Obtain and use potable water for hydrostatic pressure testing.
4. Both sustained pressuring testing and leakage testing required, performed simultaneously, if possible.
5. Both sustained pressure testing and leakage testing shall have a minimum duration of 2 hours, excluding any conditioning period of the system.
6. Testing pressure: 150 psi or 1.5x the system operating pressure, whichever is greater.
   a. Testing pressure to be achieved at the highest point of the piping segment, unless maximum pressure at the lowest point exceeds pipe pressure class rating.
7. All equipment information shall be provided to and approved by the COR prior to testing.
8. Calibrated flow meter for measuring makeup water volume required.
   a. Makeup water allowances shall meet the requirements of AWWA C604.
   b. Verify and record allowance calculations.

c. Verify and record makeup water volumes required.

9. Calibrated pressure gauges used during hydrostatic pressure testing to verify pressure throughout pipe.
   a. Locations:
      1) Test pump.
      2) High point in piping system.
   b. Calibrated for accuracy and dial range within 12 months of hydrostatic pressure testing.
10. Verify and record successful weld, joint, and gauge operations during hydrostatic pressure testing.
11. Test failures include:
   a. Pressure drop acceptable tolerances include pressure changes within instrument resolution.
   b. Leakage exceeding allowance calculations.
   c. Weld, joint, gauge, and/or appurtenance failing during testing.
12. Retesting:
   a. Retesting of the hydrostatic pressure and leakage tests is required upon any test failures.
   b. Retesting shall be done until tests are successfully completed.

D. Post-Repair Flush:
1. Required only if repairs to piping segments are necessary.
2. Obtain and use potable water for flushing.
3. Minimum velocity in accordance with AWWA C651.

E. Disinfection.
1. Prior to Disinfection, complete Initial Flushing.
2. Obtain and use potable water for disinfection.
3. In accordance with AWWA C651.
   a. Introduce water into pipeline system including backflow prevention.
   b. Perform disinfection procedures.
   c. Minimum initial free chlorine concentration in accordance with AWWA C651 throughout system.
   d. Operate valves and hydrants to ensure these are disinfected.
   e. Maintain minimum free chlorine concentration throughout the system for a contact time in accordance with AWWA C651 based on method chosen.

F. Final Flushing.

1. Prior to Final Flushing, complete Disinfection.
2. Obtain and use potable water for flushing.
3. Flush system as soon as possible after the completion of disinfection.
4. Dechlorinate and discharge water in accordance with AWWA C655.
5. Confirm the system is fully flushed by either:
    a. Measuring feed and discharge free chlorine concentrations and ensuring they are the same (within 0.30 mg/L of each other).
    b. Measuring a discharge free chlorine concentration less than 1.0 mg/L.

G. Verification.

1. Bacteriological Testing.
    a. In accordance with AWWA C651.
    b. Total coliform, Escherichia coli, and heterotrophic plate count (HPC) bacteria testing required.
    c. Use the approved testing laboratory.
    d. Samples to be collected within 24 hours of completing Final Flushing.
    e. Samples to be stored correctly prior to delivery to testing laboratory to prevent inaccurate analyses.
    f. Samples shall be delivered to testing laboratory within 24 hours of collection.
2. Sampling Locations.
    a. In accordance with AWWA C651.
    b. If sampling location required spacing (e.g. every 1200 ft.) cannot be obtained through existing infrastructure, install testing saddles.
3. Bacteriological Testing Results.
    a. Passing test.
        1) In accordance with AWWA C651.
        2) Meet required chlorine residual at sampling point.

H. Completion:

1. Replace permanent system devices removed for flushing, testing, disinfection, and verification activities.
2. If sampling taps installed, seal taps and include in Final Drawings.
3. Obtain written permission from COR prior to connecting to existing system.

4. Final acceptance will be approval of RSN 33 13 00-2, Flushing, Testing, Disinfection, and Verification Report.

3.05 REDISINFECTION

A. Repeat disinfection, final flushing, and verification steps if:

1. Water quality testing fails to meet requirements.
2. A disinfected portion of the system was exposed to any potential or actual contamination.
3. Water is in disinfected pipe for greater than 30 days before the start of operations and potable water delivery to customers.
    a. COR shall be notified once pipeline is disinfected to begin tracking of 30-day timeline.

B. If more than three consecutive test failures occur, COR shall be notified, and a corrective action plan shall be submitted.

END OF SECTION

SECTION 35 22 14
VALVES AND EQUIPMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Valves and Equipment:

1. Payment: Lump sum prices offered in Price Schedules for various types and sizes of valves and equipment provided as shown on drawings.

1.02 DEFINITIONS

A. Details of valves and equipment are shown on drawings: 1695-D-62548 through 1695-D-62552 and 1695-D-62558 through 1695-D-62568.

1.03 REFERENCE STANDARDS

A. American Society of Mechanical Engineers (ASME)

1.	ASME B16.5-2025	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch

B. ASTM International (ASTM)

1.	ASTM A216/A216M-21	Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
2.	ASTM A536-24	Ductile Iron Castings

C. American Society of Safety Engineers (ASSE)

1.	ASSE 1063-21	Air Valve and Vent Inflow Preventer

D. American Water Works Association (AWWA)

1.	AWWA C207-23	Steel Pipe Flanges for Waterworks Service, Sizes 4 In. through 144 In.
2.	AWWA C504-23	Rubber-Seated Butterfly Valves
3.	AWWA C507-23	Ball Valves, 6 In. Through 60 In. (150 mm Through 1,500 mm)
4.	AWWA C508-25	Swing-Check Valves for Waterworks Service, 2-In. through 48-In. (50-mm Through 1200-mm) NPS
5.	AWWA C509-23	Resilient-Seated Gate Valves for Water Supply Service

6.	AWWA C512-25	Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service
7.	AWWA C514-21	Air Valve and Vent Inflow Preventer Assemblies for Potable Water Distribution System and Storage Facilities
8.	AWWA C515-20	Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
9.	AWWA C530-22	Pilot Operated Control Valves

E. Manufacturers Standardization Society of the Valves and Fittings Industry (MSS)

1.	MSS SP-110-2010	Ball Valves Threaded, Socket–Welding, Solder Joint, Grooved, and Flared Ends

F. National Sanitation Foundation International / American National Standards Institute (NSF/ANSI)

1.	NSF/ANSI/CAN 61-24	Drinking Water System Components – Health Effects
2.	NSF/ANSI/CAN 372-24	Drinking Water System Components – Lead Content

1.04 SUBMITTALS

A. Submit the following in accordance with the requirements of Sections 01 33 00 – Submittals.

B. RSN 35 22 14–1, Commercial Products Data:

1. For items in Part 2 Products.
2. Manufacturer's technical product data, including the applicable size, standards, bill of materials, and pressure rating:
    a. For pilot-operated control valves, include the following for each type and size of valve:
        1) Parts identification lists.
        2) Lists of special tools and accessories.
        3) Schematics showing pilot-operated control system, connections and manufacturer recommended componentry:
        4) Instructions for installing, adjusting, operating, lubricating, and maintaining equipment.
        5) Cavitation chart showing flow rate, differential pressure, valve characteristic curve (Cv versus percent open), and system velocity.

C. RSN 35 22 14-2, Field Installation Procedures.

1. For items listed in Part 2 Products.
2. Pilot-operated control valves: Include circuit schematic of pilot-operating system.

D. RSN 35 22 14–3, Shop Testing Certifications, Reports, and Data:

1. Hydrostatic pressure testing certifications and pressures used in testing valves:
   a. AWWA gate valves tested in accordance with AWWA C509, or AWWA C515.
   b. Pilot-operated control valves tested in accordance with AWWA C530.
   c. Silent check valves tested in accordance with AWWA C508.
2. Affidavit of compliance to applicable standards:
   a. AWWA gate valves.
   b. AWWA silent check valves.
   c. Pilot-operated control valves.
3. Include pertinent observations and notable occurrences during testing.

E. RSN 35 22 14–4, Service Manuals and Final Data:

1. Items in Part 2 Products.
2. Complete parts identification lists including clearly identified recommended spare parts, lists of special tools and accessories, detailed instructions for installing, adjusting, operating, lubricating, and maintaining the equipment.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Take precautions during loading, transportation, unloading, storage, and laying, to prevent damage to the valves and equipment:

1. Repair damage to the valves and equipment as directed if, in the opinion of COR, a satisfactory repair can be made; otherwise replace damaged valves and equipment.
2. Ship, handle, store, and position according to manufacturer's instructions.
3. Cover valves until installed.

1.06 LOCATION DATA

A. Refer to table as shown on drawing 1695-D-62558 for flow control valve sizes.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Valves and equipment materials:

1. ANSI/NSF/CAN 61 certified.
2. ANSI/NSF/CAN 372 certified lead-free.
3. Suitable for potable water.

B. Valve Body Flange Faces and Flanges:

1. Valve manufacturers are responsible for providing the correct flange face gasket surface finish.
2. Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for the gasket material provided and for the operating pressures.
3. Valve flange bolt patterns to match pipe flanges.

C. Furnish valves and actuators of same type and size from same manufacturer.

D. Spare parts:

1. Provide one extra set of seats for each size and type of gate valve.
2. parts for each size and type of air valve including:
    1) Seats, lever, and gasket.
    2) Additional parts recommended by manufacturer.
    3) Box and clearly label box: Include list of spare parts inside box.
3. Store spare parts in location as directed by COR.

2.02 VALVES AND OPERATOR

A. Pilot Operated Control Valves with Solenoid Shut-off:

1. General:
    a. AWWA C530.
    b. Full, globe style.
    c. Anti-cavitation trim as recommended by manufacturer.
    d. During valve operation, water vented from upper pressure chamber of main valve (when main valve is opening) shall be directed into pilot piping so that it discharges into downstream side of main control valve.
    e. Diaphragm operated.
    f. Pressure gauges integral to the main control valve to measure upstream and downstream pressures.

g. Drip tight shut-off.

h. Sealing disk: Suitable for potable water.

i. ASTM A536 ductile iron or ASTM A216 WCB cast steel.

j. Stainless steel trim.

k. Control system components to be manufactured by same company as main valve.

l. Pilot piping for the valves:

1) Design, furnish, and test per manufacturer's recommendations.

2) Hydraulic (water) operated control pilots with seamless copper or stainless steel tubing.

3) Heavy-duty, ball-type shut off valves.

4) Opening and closing speed control valves.

5) Pilot "Y" strainers with blowdown valves.

6) Check valves.

7) Atmospheric drain.

8) Pressure gauges.

9) Provide automatic vent valves.

m. Valve visual position indication:

1) Stainless steel position indicator rod.

2) Brass or stainless steel protective housing which allows viewing of valve position from two sides.

3) Manual air purge valve at top of assembly.

4) Pyrex tube or covers to enclose stem inside housing.

n. Furnish and install pilot piping and associated valving and fittings as recommended by the valve manufacturer.

2. Model 106-SC manufactured by Singer Valve, www.singervalve.com, Model No. 43-01, manufactured by Cla-Val Company, www.cla-val.com, Model A115 Series manufactured by Apollo Valves, www.apollovalves.com, Model 120-01 manufactured by OCV Control Valves, www.controlvalves.com; or equal, having the following essential characteristics:

a. Automatically controls flow through valve to specified flow rate.

b. Flow control settings shall be field adjustable.

c. Orifice plates:

1) Install orifice plates and valves as shown on the drawings. Actual valve and orifice plate locations may vary depending on control valve manufacturer's recommendations.

2) Provide orifice plate bore size in accordance with manufacturer's recommendations to meet required flow rates.

d. Solenoid shut-off valve:

1) 120 vac.

2) Control valve to close when supplied power.

3) Control valve will remain open on loss of power.

e. Limit switch:

1) Adjustable.

2) Integral to control valve.

3) Actuates when valve is fully closed.

4) Double pole double pole switches rated at 1 amp at 125 vdc and 120 vac.

5) UL rated with NEMA 4 enclosure.

B. Exposed Ball Valves:

1. Carbon Steel, Cast Iron or Bronze:

2. 3-inch and smaller:

a. 70LF-100 Series manufactured by Apollo Valves, www.apollovalves.com; or equal, having the following essential characteristics:

1) Cold water pressure rating of not less than 300 psig

2) Body: Full ported carbon steel or lead-free bronze.

3) Ball: Stainless steel or lead -free brass.

4) Valve stem: Stainless steel or lead-free brass.

5) Packing gland: Steel or brass.

6) Install where shown on drawings. Bronze or carbon steel ball valves shall be installed on carbon steel pipe.

7) Bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.

8) MSS SP–110.

9) Female threaded ends.

3. 4-inch and larger:

a. Val-Matic Rubber Seated Ball Valves, manufactured by Val-Matic, www.valmativ.com; or equal, having the following essential characteristics;

1) Flanged ends:

a) AWWA Class 150 valves: Flanges suitable for connection to AWWA Class D flanges or ANSI Class 150 flanges.

2) Valves suitable for manual service.

3) AWWA C507.

4) Pressure ratings suitable for steel pipe pressure.

4. Provide dielectric insulating connections as necessary to prevent galvanic corrosion between dissimilar metals.

C. Buried Ball Valves (Curb Stop):

1. Bronze:

2. 3-inch and smaller:

a. Model 76171 manufactured by A.Y. McDonald, www.aymcdonald.com; or equal, having the following essential characteristics:

1) Cold water pressure rating of not less than 300 psig

2) Body: Full ported carbon steel or lead-free bronze.

3) Ball: Stainless steel or lead -free brass.

4) Valve stem: Stainless steel or lead-free brass.

5) Packing gland: Steel or brass.

6) Install where shown on drawings. Bronze or carbon steel ball valves shall be installed on carbon steel pipe.

7) Bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.

8) Female threaded ends.

D. Gate Valves:

1. Series A-2361 manufactured by Mueller Co. LLC, www.muellercompany.com, High Pressure Resilient Wedge Gate Valve manufactured by Kennedy Valve, www.kennedyvalve.com; or equal, having the following essential characteristics:

a. AWWA C509 or AWWA C515.

b. OS&Y or NRS

c. Ductile iron body and wedge shall be made of ductile iron.

d. Bonnet: Ductile iron or cast iron.

e. Handwheel.

f. Valve ends shall be flat faced flanges and shall match the drilling dimensions and pattern of AWWA C207 Class F flanges.

g. Wedges shall be fully encapsulated with EPDM.

h. Minimum cold-water working pressure rating of 300 psig.

i. Zero leakage.

E. Check Valves:

1. Series 581 Silent Check Valve manufactured by Cla-Val, www.cla-val.com, or equal, having following essential characteristics:

a. Silent operating type.

b. Globe style.

c. Center guided, spring loaded disc.

d. All parts to be field replaceable.

e. Resilient seal.

f. Zero leakage.

g. Minimum working pressure rating of 300 psig, cold water service.

h. Suitable for bolting to AWWA Class F steel pipe flanges.

i. Valve body: ASTM A536 grade 65-45-12 ductile iron or ASTM A216 grade WCB cast steel.

j. Disc: ASTM A351 grade stainless-steel.

k. Seat and compression spring, type 316 stainless-steel.

l. Valve body with integral flanges (not wafer style).

m. Flat face flanges: Provide spiral or concentric serrated grooved flange face gasket surface finishes that are suitable for flat gasket material.

F. Combination Air Valves:

1. Manufactured by DeZurik/Apco, www.dezurik.com; Cla-Val, www.cla-val.com; Val-Matic Valve and Manufacturing Corporation, www.valmatic.com; or equal having the following essential characteristics:

a. Cold water service.

b. Cold-water pressure rating: 300 psig, minimum.

c. Single or dual body.

d. Body material: Ductile iron or cast steel.

e. AWWA C512.

f. Actuated by a float or floats.

g. Stainless steel internal parts such as guides, bushings, and screws.

h. Stainless steel floats.

i. Float-pivot supports: Brass, bronze, stainless steel, or cast iron.

j. Inlet: Threaded

k. Provide combination air-inlet and air-release type or air and vacuum type actuated by a float as shown on the drawings.

l. Remains open for filling line until water has displaced air at point of attachment of valve assembly after which it closes.

m. Opens when pressure in pipeline drops sufficiently to create a vacuum.

n. Releases trapped air when the pipeline is under pressure.

o. 2-inch Air release minimum orifice size: 1/16-inch.

p. 3-inch Air release minimum orifice size: 5/64-inch

q. Operating pressure for seat materials: Resilient seat suitable for 0 to 300 psig.

r. Furnish with ball valves in accordance with paragraph 2.01 F:

1) One ball valve on top, permits checking effectiveness of air valve.

2) One ball valve on bottom, allows valve to be drained.

3) For dual body valves, provide full port isolation ball valve between main body and air release valve.

G. Inflow Preventer Assembly: Val-Matic model 1302/1303 with optional bracket SPK-1302/1303-80, and field test kit SPK-1302/1303 as applicable per air valve size, manufactured by Val-Matic Valve and Manufacturing Corporation, 905 Riverside Drive, Elmhurst, IL 60126; or equal, having the following essential characteristics:

1. Manufactured in accordance with AWWA C514 and ASSE 1063.

2. Stainless steel internal parts such as guides, bushings, and screws.

3. Upper and lower chambers: Ductile iron, ASTM A536 Grade 65-45-12.

4. Basket strainer:

a. Type 304 stainless steel.

b. 24# mesh opening (0.0277 inch).

c. Quick disconnect joint.

d. Total flow area of at least four times circular cross-sectional area having a diameter equal to nominal size of air valve outlet.

5. Float checks and trim: Type 316 stainless steel.

6. Cold-water pressure rating for body: Minimum of 25 psi.

7. Resilient seat: EPDM with fiberglass reinforcement.

8. Inlet connection: Internally threaded NPT.

9. Piping manifold and wall bracket as shown on drawings.

10. Supply mounting bracket and field test kit with all appurtenant piping and valves per air valve as shown on drawings.

H. Butterfly Valves:

1. Cold water pressure up to 150 psig: AWWA C504, Class 150B.
2. Cold water pressures from 151 psig up to 250 psig: AWWA C504, Class 250B.
3. Bubble tight at rated pressure in either direction and suitable for operation after long periods of inactivity.
4. Tested in accordance with AWWA C504 including performance, leakage, and hydrostatic testing.
5. Axis of valve leaf: Horizontal when valve is in installed position except where shown on drawings.
6. Flanged ends:
   a. Class 150B valves: Body flanges suitable for connection to AWWA Class D flanges.
   b. Class 250B valves: Body flanges suitable for connection to AWWA Class F flanges.
7. Valve Seat:
   a. Rubber seat: Clamped, or mechanically secured to body or disc.
   b. Material: AWWA C504, except bonded rubber seats will not be accepted.
   c. Field replaceable seat.
   d. Located either in the valve body, mating with a stainless steel seating surface, located on valve disc or in valve disc, mating with a stainless steel seating surface located in valve body.
8. Manual Operators:
   e. Manually operated butterfly valves with worm-gear or traveling nut type manual operator in accordance with AWWA C504.
   f. Ensure rim pull and component strength meets AWWA C504.
   g. Operators that open and close valves at full unbalances pressure rating of valve.
   h. Gears of self-locking type; holds valve in all positions without fluttering or creeping.
   i. Handwheel where shown on drawings.
9. Buried butterfly valves:
   a. Valve boxes.
   b. Square nut operator: 2-inch.
   c. Provide a total of 8 steel tee-handle wrenches, with a length of 4-feet.
      1) Torque rating Match or exceed maximum torque input of valve actuator.

2) Tee wrench:

a) Capable of withstanding 2 times the torque rating.

b) Painted with primer and topcoat to manufacturer's standard.

d. Steel extension stems:

1) Required to provide operating nut to within 6-inches of ground surface for buried valves:

2) Rating to match or exceed maximum torque input of valve actuator.

3) Steel centering ring with 1/8-inch minimum thickness and welded all around to top of extension stem just below the operating nut.

4) Bottom socket with mud vents, welded or pinned to provide a secure connection to the square steel extension stem.

5) Connect top nut to square steel stem with a pin or two cap screws. Capable of withstanding 2 times the torque rating.

6) Painted with primer and topcoat to manufacturer's standard.

I. Valve Boxes:

1. Model 6855 series, manufactured by Tyler Union, www.tylerunion.com; or equal, having the following characteristics:

a. Cast iron locking lid with pentagon shaped bolt head.

b. Cast iron extension from valve nut operator to ground surface.

c. Cast iron collar with lid marked "WATER".

d. 5 1/4-inch minimum inside diameter cast iron.

e. 2-piece or 3-piece and slip type.

2. Base and adequate extension items to extend from valve nut operator to ground surface.

3. Refer to Detail A on drawing 1695-D-62549.

J. Air Vent Damper:

1. Model No. 1512, FrostSafe two-way damper manufactured by Val-Matic Valve and Manufacturing Corporation, 905 S. Riverside Drive, Elmhurst IL 60126; or equal having the following essential characteristics:

a. For 6-inch air vent as shown on drawing 1695-D-62551.

b. Suitable for connection between AWWA Class D flanges.

c. Allows 2-way flow.

d. Minimizes thermal exchange of cold and warm air in and out of valve vault.

e. Allows vault to vent at full air valve capacity.

K. Pressure Reducing Valve:

1. Single stage and suitable for water.
2. End connections: Threaded for regulators NPS 2-inch and smaller
3. Minimum working pressure: 300 psi.

L. Hose fittings:

1. Hose Bibb:
   a. Operation: 1/4 turn.
   b. End connections: Threaded.
   c. Suitable for working pressure of pipe.
   d. Hose adapter;End connections: Threaded to hose barb.
   e. Suitable for working pressure of pipe.
2.

M. Magnetic Flowmeter:

1. Electromagnetic Flowmeter, Ultra Mag, as manufactured by McCrometer,; or equal, having the following essential characteristics;
   a. Accuracy: Plus, or minus, 0.5 percent of all flowrates, over range of measurement.
   b. +0.5% Accuracy over specified flow range:
   c. Straight pipe installation clearances:
      1) 3 diameters upstream.
      2) 2 diameters downstream.
   d. Suitable for operation in ambient temperatures ranging from -4 to 140 degrees F.
   e. Flanged, cylindrical flow-tube with polyurethane liner, grounding rings and gaskets.
   f. Flanges:
      1) Suitable for bolding to AWWA Class F.
   g. Sensor Tubes:
      1) Stainless steel electrode material
      2) Bonded epoxy liner.

h. Transmitter:

1) Modbus or 4-20mA HART.

2.03 APPURTENANCES:

A. Gaskets:

1. Multi-Swell Style 3760 manufactured by Garlock, www.garlock.com; or equal with the following essential characteristics:
   a. Minimum 300 psig cold water pressure rating.
   b. AWWA C207 or ASME B16.5.
   c. Flat, full face type.
   d. Suitable for cold water service.
2. Do not use compressed gaskets.

B. Delivery Point Signage Tags:

1. As shown on Drawing 1695-D-62568.
2. Engraved metal tag (blue with white lettering).
3. Wire Rope (1/8" diameter).
   a. Provide 6 spare tags, wire rope and crimps.

2.04 CONTRACTOR SOURCE QUALITY TESTING

A. Shop Testing:

1. Test valves at manufacturer's facility or an alternate facility approved by COR.
2. Tests will be Government witnessed. Notify COR 30-days prior to valve testing.
3. Minimize number of Government trips by testing valves from same manufacturer during same Government trip.
4. Prior to shop leakage test and in presence of Government, Contractor shall verify that when valve is fully closed, valve position indicator indicates, "CLOSED".
5. If indicator is not in closed position (exactly at closed), remove operator from valve, fully close valve, reinstall operator so that it indicates, "CLOSED".
6. Shop leakage test acceptance:
   a. Spray or high velocity leakage will not be acceptable.
   b. If valve leaks, adjustments shall be made until valve is drip tight with no leakage. Adjustments may include adjusting valve stops, seats, actuators, or other components.
   c. If valve is not drip tight, valve is unacceptable.

d. After any adjustments are made, valve position indicator shall be inspected again and corrected if necessary.

7. Manually operated gate valves:

a. Leakage test in both directions in accordance with AWWA C509 or AWWA C515, as applicable. Record results.

b. Hydrostatic test in accordance with AWWA C509 or AWWA C515, as applicable. Record pressures used.

c. Pressure test valves and their seats for leaks with water in 100 percent closed position for a timed period of 5-minutes with pressure equal to rated design pressure. Valves shall be drip tight with no leakage.

d. Downstream position of valve shall be visible to Government inspector.

8. Check valves (globe style) shop testing:

a. AWWA C508 seat leakage and valve body hydrostatic test.

b. In accordance with manufacturer's recommendations.

9. Pilot-operated control valves with solenoid shutoff shop testing:

a. AWWA C530 seat leakage test and valve body hydrostatic test.

b. Manufacturers' requirements.

PART 3 EXECUTION

3.01 INSTALLATION

A. Dielectrically insulate between dissimilar metals:

1. Provide insulating type unions at threaded valve, pipe or fitting connections between dissimilar metals.

B. Provide a sufficient quantity of unions for removal, replacement and maintenance of valves and equipment.

C. Install the valves and equipment to line and grade as shown on drawings.

D. Install valves with flanged connections so that the bolt holes straddle the vertical centerline when in the installed position.

E. Install valves and gaskets in accordance with the manufacturer's instructions.

F. Valve Tags:

1. Provide brass or stainless steel tags for all valves identified in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

2. Stamp or engrave tag with the valve identification code designated in the Piping and Instrumentation Diagram drawings 1695-D-T0152 through 1695-D-T0161.

3. Permanently attach tag to valve with wire and metal crimp-type connector.

3.02 CONTRACTOR FIELD QUALITY TESTING

A. Field Pressure Testing:

1. After each valve has been completely installed, test valve in place by opening and closing valve through full range of operation 3 times.
2. Furnish testing equipment, as approved by COR, which provides observable and accurate measurements of leakage under specified conditions.
3. Maximum Filling Velocity: 0.25 feet per second calculated based on full area of pipe.
4. Expel air from piping system during filling.
5. Test Pressure:
   a. 150 percent of steel manifold pipe normal operating pressure.
   b. Maximum test pressure not to exceed 300 psi.
   c. Test pressure for some segments may be reduced subject to COR approval.
6. Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached.
7. Maintain hydrostatic test pressure continuously for 2 hours minimum.
8. Maximum Allowable Leakage: No leakage.
9. Dispose of water used in testing.

B. Test pilot operated control valve per manufacturer's recommendation.

1. Set flow control valve full closure speed to a minimum of 60 seconds, or as close as allowed by manufacturer recommendation.

C. Test Flow control valve and Flowmeter per specification Section 01 80 15 - Commissioning.

3.03 COATINGS

A. Coat valves and equipment in accordance with Section 09 96 20 – Coatings.

END OF SECTION

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SECTION 52 00 00
DRAWINGS

PART 1 GENERAL

1.01 DISCREPANCIES, ERRORS, OR OMISSIONS

A. Inform the CO of discrepancies discovered on drawings in accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction.

1.02 PROJECT CONDITIONS

A. Where there are differences as determined by the CO between details and dimensions shown on drawings and details and dimensions of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.03 INFORMATION DRAWINGS

A. Drawings marked "For Information Only" in the drawing list are included to show existing features about which knowledge is required to perform work under this contract. These drawings do not show work to be performed under this contract.

B. If there are differences as determined by the CO between details and dimensions shown on these drawings and those of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.04 STANDARD DRAWINGS

A. Standard drawings may show details which are not a part of work under this contract. Disregard details shown on these drawings which are not applicable to work under this contract.

1.05 DRAWING NUMBERS IN NUMERICAL ORDER

A. Specification drawings are listed in numerical order in Table 52 00 00B - Drawings in Numerical Order.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-	
1695-D-62526 Rev1	***1***
1695-D-62527	10
1695-D-62528 Rev1	***11***
1695-D-62529 Rev1	***12***
1695-D-62530	13
1695-D-62531 Rev1	***14***
1695-D-62532 Rev1	***15***
1695-D-62533 Rev1	***16***
1695-D-62534 Rev1	***17***
1695-D-62535 Rev1	***18***
1695-D-62536 Rev1	***19***
1695-D-62537 Rev1	***20***
1695-D-62538 Rev1	***21***
1695-D-62539 Rev1	***22***
1695-D-62540 Rev1	***23***
1695-D-62541 Rev1	***24***
1695-D-62542 Rev1	***25***
1695-D-62543 Rev1	***26***
1695-D-62544 Rev1	***27***
1695-D-62545 Rev1	***28***
1695-D-62546	29
1695-D-62547	30
1695-D-62548	31
1695-D-62549 Rev1	***32***
1695-D-62550	33
1695-D-62551	34
1695-D-62552	35

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-62553	36
1695-D-62554 Rev1	***37***
1695-D-62555	38
1695-D-62556	39
1695-D-62557	40
1695-D-62558 Rev1	***41***
1695-D-62559	42
1695-D-62560	43
1695-D-62561 Rev1	***44***
1695-D-62562 Rev1	***45***
1695-D-62563 Rev1	***46***
1695-D-62564 Rev1	***47***
1695-D-62565 Rev1	***48***
1695-D-62566 Rev1	***49***
1695-D-62567 Rev1	***50***
1695-D-62568 Rev1	***51***
1695-D-62569	52
1695-D-62570	53
1695-D-62571 Rev1	***54***
1695-D-62572 Rev1	***55***
1695-D-62573 Rev1	***56***
1695-D-62574 Rev1	***57***
1695-D-62575 Rev1	***58***
1695-D-62576 Rev1	***59***
1695-D-62577 Rev1	***60***
1695-D-62578	61
1695-D-62579	62
1695-D-62580	63
1695-D-62581	64

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-62582	65
1695-D-62583	66
1695-D-62584	67
1695-D-62585	68
1695-D-62586	69
1695-D-62587 Rev1	***70***
1695-D-62588	***71***
1695-D-62590	73
1695-D-62591	74
1695-529	
1695-529-60481	2
1695-529-60488	3
1695-529-60489	4
1695-529-60490	5
1695-529-60491	6
1695-529-60492	7
1695-529-60493	8
1695-529-60494	9
Informational	
085-N-1110	97
085-N-1111	98
085-N-1115	99
085-N-1116	100
085-N-1120	101
085-N-1121	102
085-N-1125	103
085-N-1126	104
085-N-1130	105
085-N-1131	106

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
085-N-1135	107
085-N-1136	108
085-N-1150	109
085-N-1151	110
1695-D-60119	75
1695-D-60129	76
1695-D-60150	77
1695-D-60157	78
1695-D-60253	79
1695-D-60260	80
1695-D-60281	***114A***
1695-D-60282	***114B***
1695-D-60283	***114C***
1695-D-60296	***114D***
1695-D-60297	81
1695-D-60298	***114E***
1695-D-60309	82
1695-D-60310	***114F***
1695-D-60311	***114G***
1695-D-60322	83
1695-D-60324	84
1695-D-60325	85
1695-D-60326	86
1695-D-60327	87
1695-D-60684	88
1695-D-60685	***114H***
1695-D-60686	***114I***
1695-D-60695	89
1695-D-60696	***114J***

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
1695-D-60697	***114K***
1695-D-60713	90
1695-D-60720	91
1695-D-60722	92
1695-D-60724	93
1695-D-60727	94
1695-D-60728	95
1695-D-60731	96
6922268 C-1	112
6922268 FIG-1	114
6928904 C-1	113
6928997 C-1	111
40-D and 104-D	
104-D-254	128
104-D-757	129
104-D-1150	130
104-D-1165	131
40-D-4334	115
40-D-4335	116
40-D-4753	117
40-D-5246	118
40-D-5247	119
40-D-6234	120
40-D-6551	121
40-D-6552	122
40-D-7102	123
40-D-60001	124
40-D-60002	125

Table 52 00 00B – Drawings in Numerical Order

Drawing No.	Sheet No.
40-D-60003	126
40-D-60004	127

END OF SECTION

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SECTION 53 20 00
RECORDS OF GEOLOGIC AND SUBSURFACE INVESTIGATIONS

PART 1 GENERAL

1.01 GEOLOGY DRAWINGS

A. Some drawings are in color to highlight geologic information. Geologic information shown on drawings is of equal importance regardless of whether or not it is highlighted.

Geology Drawings

Sheet No.	Drawing No.	Title
G1	103-D-347	Unified Soil Classification – Including Identification and Description
G2	40-D-7022	Geology for design & Specifications, Standard Descriptors and Descriptive Criteria for Rock
G3	40-D-7023	Geology for design & Specifications, Standard Descriptors and Descriptive Criteria for Discontinuities
G4	1695-529-60459	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, General Vicinity Map
G5	1695-529-60463	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, General Geologic Legend, Explanation and Notes
G6	1695-529-60464	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Sanostee Turnout, Surface Geology & Location of Exploration
G7	1695-529-60465	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Newcomb Turnout, Surface Geology & Location of Exploration
G8	1695-529-60466	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Naschitti Turnout, Surface Geology & Location of Exploration
G9	1695-529-60467	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, BBN9 Turnout, Surface Geology & Location of Exploration

Geology Drawings

Sheet No.	Drawing No.	Title
G10	1695-529-60468	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Coyote Canyon (BBN9) Turnout, Surface Geology & Location of Exploration
G11	1695-529-60469	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Rock Springs Turnout, Surface Geology & Location of Exploration
G12	1695-529-60470	Navajo Gallup Water Supply Project, New Mexico, San Juan Lateral, Turnout Buildings, Window Rock Turnout, Surface Geology & Location of Exploration

1.02 GEOLOGIC LOGS

Geologic Logs

Sheet No.	Title
Test Pits	
G13	Log of Test Pit No. TPTB-25-1
G14	***Log of Test Pit No. TPTB-25-2***
G15	Log of Test Pit No. TPTB-25-3
G16	Log of Test Pit No. TPTB-25-4
G17	Log of Test Pit No. TPTB-25-5
G18	Log of Test Pit No. TPTB-25-6
G19	Log of Test Pit No. TPTB-25-7
G20	Log of Test Pit No. TPTB-25-8
G21	Log of Test Pit No. TPTB-25-9
G22	Log of Test Pit No. TPTB-25-10
G23	Log of Test Pit No. TPTB-25-11

Geologic Logs

Sheet No.	Title
G24	Log of Test Pit No. TPTB-25-12

1.03 LABORATORY TEST DATA

Laboratory Test Data

Sheet No.	Title
Laboratory Test Data for Test Pits	
G25	Summary of Physical Properties Test Results, Navajo Water Supply Project, Turnout Buildings
G26-G42	***Soil Corrosivity Testing***

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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